On Information, Compression and Generalisation

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Abstract

We work towards an information-theoretic description of generalisation. We begin by introducing the foundations of information theory and exploring the nature of information. We underline the equivalence between compression, prediction and information by studying the limits of coding efficiency. We use this understanding to justify and derive our information-based approach to learning.

Declaration

I declare that this thesis was composed by myself and that the work contained therein is my own, except where explicitly stated otherwise in the text.

(*Redmond Bowcott*)

To my supervisor, Dr. Joan Simon, for introducing me to the beauty of information theory and for all of our discussions this year.
To Dhruv, Jonah, Jonny, Piotr and the many others who have shared their passion for mathematics with me and who I am grateful to have learnt alongside.
To Ethan and Niranjan, for their advice and encouragement.
To my flatmates - David, Louis, OJ and Shiv - and friends, for their kindness and humour.
And most of all to my family, for their love and support.

This project report is submitted in partial fulfilment of the requirements for the degree of *BSc Mathematics (Hons)*.

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Chapter 1

Introduction

1.1 Information

We gain information by observing events whose outcomes are uncertain to us. Before defining the measures of information with which we shall work, let us first look to build an intuitive characterisation of it.

To begin, consider the two-player game of Guess Who. The rules are as follows: at the start of the game, each player independently and randomly selects one of 24 characters. They then take it in turns to ask questions, usually about the appearance of the other's chosen character - for example their hair colour, or whether they wear glasses - with only a 'Yes' or a 'No' allowed as response. The winner is the one who can guess their opponent's character first. For any given physical property, there are 5 of the 24 with that property, and 19 without.

For a player, we can frame this probabilistically as follows: the identity i of your opponent's character is a uniformly distributed random variable that can take on 24 values. Each question about a characteristic is deterministic given i. However since we do not know i, we can view questions as random variables themselves. For any such question, the probability of a 'Yes' is $\frac{5}{24}$ and the probability of a 'No' is $\frac{19}{24}$ when not conditioning on any prior questions. Although we deal with a binary random variable here, we hope that the intuitions fairly naturally extend to discrete random variables more generally.

First, let us think about how informative different outcomes of a question should be.

Example 1.1. Suppose you are playing a friend at Guess Who. For your first question, you ask whether their character has brown hair. How informative should a 'Yes' be in response, and how informative should a 'No' be in comparison?

The level of informativeness is known as the information content:

Definition 1.1 (Information Content)**.** The information content of a realisation of a random variable is a measure of the information gained from that particular observation.

Hearing a 'Yes' immediately narrows down the possible number of characters to 5; hearing a 'No' leaves us with 19. So we have gained more information about the opponent's character by hearing a 'Yes': a 'Yes' specifies a smaller section of the sample space than a 'No' does. Therefore the information content of hearing a 'Yes' should be greater.

This insight leads to two fundamental properties of information:

Proposition 1.1 (Likelihood Determines Information Content)**.** *The probability of a particular outcomes determines how much information we gain from observing it.*

Proposition 1.2 (Less Likely Outcomes have a Higher Information Content)**.** *For outcomes a, b with probabilities $p_a < p_b$, neither 0, the information content of observation a should be higher than that of observation b.*

Any sequence of questions - more specifically any sequence of corresponding answers - specifying your opponent's character has received the same total information content, since the sequence has uniquely identified one character out of 24. The aim of Guess Who is to work out your opponent's character as quickly as possible. Since we cannot choose an answer, we should look to maximise the information content in expectation at each step.

Definition 1.2 (Entropy)**.** We call the expected information content of an observation of a random variable its entropy.

Following on from Proposition 1.1, we can infer the following:

Proposition 1.3 (Entropy is a Function of the Distribution)**.** *Since information content is determined by the probability of an outcome, entropy must be determined by the distribution of the random variable.*

To make this slightly stronger:

Proposition 1.4 (Entropy is Only a Function of the Distribution)**.** *The entropy does not depend on the values taken by the random variable. It is only a function of the random variable's distribution.*

So how does entropy change across a distribution?

Example 1.2. In the traditional Guess Who setup, cards for each character are placed on a 6x4 board. As you receive answers to questions, you flip down the cards which do not fit the profile for your opponent's character.

Suppose you consider bending the rules of Guess Who slightly. You have to decide between asking whether their character has brown hair, or whether their character lies on the left side of the board. Which question should you ask?

The latter pares down the space equally with either outcome, whereas the former has a chance to significantly reduce the number of possible characters.

More informative events are less probable, so perhaps an initial instinct might be that the entropy of every random variable is equal: the average amount of information gained per question is the same. However consider the following:

Example 1.3. Suppose bending the rules is not allowed. You now look to decide whether to ask if their character's hair is brown, as before, or if their character is Susan - one of the 24.

In this case, it seems clear that asking about their character's hair is better. In fact, it seems hard to think of a worse approach than just guessing each character once in turn. The one type of question which is clearly worse is an entirely wasted question: one which does not eliminate or specify a single character. In line with our description of information at the start of this section, we can therefore intuit that as random variables become deterministic, their entropy must vanish. In fact considering all other possible questions, we can say something stronger:

Proposition 1.5 (Entropy Vanishes if and only if a Variable is Deterministic)**.** *For a random variable with possible realisations* $\{a, b, c, \ldots\}$*, the limit of the entropy as* $p_a \to 1$ *is 0.*

For a non-deterministic random variable, the entropy is non-zero.

Let us return to Example 1.2, though. The intuition that we should ask about hair colour instead of guessing a character suggests that entropy increases as the distribution becomes more even, but there might be a sweet spot between balancing out the probabilities and allowing for an answer which could provide a significant amount of information.

Here is a re-framing of Example 1.2 which might provide some clarity:

Example 1.4. Suppose that we index each character according to their position on the board, going column by column so that the left hand side of the board is numbered 1 to 12 and the right hand side has indices 13 to 24.

In another procedure, we could index those characters with brown hair 1 to 5 and all the rest 6 to 24.

Identifying our opponent's character can be viewed as equivalent to identifying an integer drawn uniformly at random from the set $\{1, 2, \ldots, 24\}$. Asking an initial question is equivalent to asking whether the opponent's character is less than or equal to the number of individuals with the characteristic - for example 'Does your character have brown hair?' can be translated as 'Is your randomly drawn integer less than or equal to 5?'.

For each question we can just re-index the remaining possible candidates in this manner and the analogy remains. What is the best number to ask first?

It hopefully appears clearer now that an equal division of the sample space is the optimal choice: this is the rationale behind binary search, for example. Recall that we are looking to maximise expected information at each step. The preference for a question with equal possibility of either outcome therefore suggests that a uniformly distributed random variable should have the highest expected information content.

Proposition 1.6 (Entropy is Maximised by Uniformly Distributed Random Variables)**.** *For a fixed number of outcomes, the entropy of a random variable is maximised when these random variable each have an equal probability.*

Through these results, we have already formed a fairly detailed picture of information. To recap: information is a function of the outcome of a random variable and less likely events provide more information. Entropy is associated with the informativeness of a random variable as a whole and is a function of the distribution of the variable. It is maximised by the uniform distribution and vanishes as the variable becomes deterministic.

It is worthwhile here to pause momentarily to discuss some framings of entropy. One already discussed is that of expected informativeness. Another common one is that since information content is tied to how unlikely a realisation is, entropy is a measure of the unexpectedness of a random variable. To propose one that feels more fitting for the discussion ahead:

Proposition 1.7. *Entropy is a measure of the difficulty of describing an outcome within a distribution.*

In this light, entropy is fundamentally a measure of the complexity of a distribution.

1.1.1 Shannon Entropy

The desire for a theory of information first arose in the early 1900s, as the emergence of the telecommunications industry began to allow for the rapid transmission of data. However the exact form that a measure of information should take was unclear. When in 1948 Claude Shannon published *A Mathematical Theory of Communication* [17], he introduced the following formulae to describe the information content and entropy of a random variable:

Definition 1.3 (Shannon Information Content)**.** The Shannon information content of a realisation $x \in \mathcal{X}$ is given by

$$h(x) := -\log_2 p(x).$$

Definition 1.4 (Shannon Entropy)**.** The Shannon entropy of X is given by the expected information content of an observation of X:

$$H(X) := -\mathbb{E}[h(x)] = -\sum_{x \in \mathcal{X}} p(x) \log_2 p(x).$$

He not only showed that these forms could be derived from an set of axioms, but also that they appeared naturally as a fundamental limit in the problem of data compression. In particular, the Shannon entropy of a random variable appeared as a bound on how well a variable's realisations could be compressed, or equivalently how efficiently they could be described. The nature of these results and the rigorous framework he outlined led to entropy becoming the foundation upon which the study of information would be built.

1.2 Towards Learning

At a similar time to the birth of information theory, the field of computer science, and in particular the study of computational intelligence, began to come into existence. Turing introduced his famous imitation game in the 1950 paper *Computing Machinery and Intelligence* [22], and in 1956 the Dartmouth Summer Research Project on Artificial Intelligence truly began the field - including providing it with the name of artificial intelligence. Among the guests was Claude Shannon, and among the topics of discussion even back then was neural networks [14], the architecture which underpins the language and vision models of today.

For a long time from its inception, the greatest area of focus within the field was the study of symbolic methods, which involve providing detailed sets of rules to determine the behaviour of systems. Machine learning, in contrast, concentrated on algorithms that could learn without explicit programming by observing data. In recent years, though, machine learning has become the dominant force within the field, most apparent in the focus on language modelling currently. *The Bitter Lesson* [20], written in 2019, gives the following perspective on this:

> *The biggest lesson that can be read from 70 years of AI research is that general methods that leverage computation are ultimately the most effective, and by a large margin ... the [encoding-]human-knowledge approach tends to complicate methods in ways that make them less suited to taking advantage of general methods leveraging computation.*

... The actual contents of minds are tremendously, irredeemably complex ... we should build in only the meta-methods that can find and capture this arbitrary complexity.

For nearly 50 years, Moore's Law—the observation made in 1965 by Intel co-founder Gordon Moore that the number of transistors on a microchip doubles roughly every two years—has driven exponential growth in computing efficiency [23]. The increase in processing power that this has allowed has enabled the existence of today's models by providing the scale and speed necessary to process the data required to train and run them.

The aim of machine learning is to train a model to acquire a knowledge of the underlying distribution of the data from a finite number of samples. We look to produce algorithms that can not only perform well on data already seen, but also on that unseen. The challenge is that we cannot teach an algorithm to generalise directly since by definition, generalisation must be evaluated on exactly the data that we do not have. Instead, we must balance a bias-complexity trade-off: more sophisticated algorithms can potentially learn a better approximation to the true distribution, but incur a risk of overfitting on the sample data since they have the capacity to match it more exactly.

1.2.1 An Information-Theoretic Framework

In this piece, we work towards an information-based theory of learning. We look to characterise the theoretical limits of generalisation within classes of distributions such linear or logistic models. Specifically, we characterise the rate at which a learning algorithm can approximate the true distribution within the class - the exact coefficients of the linear regression, for example - as it receives samples from the distribution. We shall arrive at two formulations of how well a class can be learnt, both of which we can motivate from our discussion so far:

1. The difficulty of learning the true distribution can be measured by how much our uncertainty of its value changes as we acquire data. Precisely, the difficulty of learning with respect to a given data-generating process can be measured by the difference between our absolute uncertainty in the true distribution and our uncertainty in the true distribution after observing the data.

   The reduction in uncertainty about one random variable upon observing another is known as the mutual information between them. Also defined by Shannon in his seminal paper, it is exactly the difference in entropy between the conditional and unconditional random variable.

2. The limit to the learnability of a class is given by the limit of the performance an algorithm can achieve when learning from data generated by a distribution in that class. The optimal learner is the one that best balances the bias-complexity trade-off.

   We can bound the sum of its bias and complexity above by the sum for any other model. At the same time, to reduce loss to a given level, a learner must have attained sufficient information about the data-generating distribution. This requires a minimum level of complexity. We can bound the learnability of the algorithm below by considering this combination of minimal complexity and corresponding loss over all possible levels of loss.

To rigorously understand these statements and this framework for learning as a whole, we first must understand the nature and behaviour of information. We shall establish how the fundamental quantities of information characterise random variables. We then turn to the limits on how efficiently information can be shared. We look to deepen the connections between the trio of information, prediction, and compression. We focus separately on lossless compression, which emphasises the natural presence of entropy, before broadening perspective and leading ourselves towards this bias-complexity trade-off by considering rate-distortion theory, which describes the limits of lossy compression.

In particular, in Chapter 2 we define the concepts of entropy, divergence and mutual information. We prove that the properties of Section 1.1 are satisfied by Shannon's theory, describe the nature of divergence and mutual information, and outline different ways these quantities can be expressed and interpreted. In Chapter 3 we prove the Asymptotic Equipartition Property, which shows that the information content of sequences concentrates in the limit (Theorem 3.1); Shannon's Source Coding Theorems (Theorems 3.2, 3.5), which identify entropy as the fundamental limit of lossless compression; and the Kraft-McMillan inequality (Theorems 3.3,3.4), which underline the connection between prediction and compression. We move towards lossy compression in Chapter 4 and focus on expanding the bounds of the last chapter by considering the limits for lossy compression (Theorem 4.1). Finally we turn to machine learning in Chapter 5. We show that an algorithm whose predictive distribution is the Bayesian posterior is the optimal learner with respect to generalisation for any underlying distribution (Lemma 5.2). We show that it learns all information that the data provides about the true distribution (Lemma 5.3), then bound it by a chosen rate-distortion function (Theorem 5.6). We complete our discussion by using these bounds to deduce that a logistic regression can be learnt at rate $\mathcal{O}(1/T)$ (Theorem 5.15).

In Appendix A, we provide a brief introduction to measure-theoretic probability. This is not necessary to understand any of the content of the remaining chapters, but provides a formal construction which should strengthen understanding of results such as the weak law of large numbers or the tower property of conditional expectation.

Chapter 2

Foundations of Information Theory

We begin by defining and characterising the fundamental measures of information with which we shall work. The definitions here follow from [1], [12] and [5].

2.1 Shannon Entropy

Consider a discrete random variable X taking values in $\mathcal{X} = \{x_1, \dots, x_n\}$ and with mass function $p(X)$.

Definition 2.1 (Shannon Information Content)**.** The Shannon information content of a realisation $x \in \mathcal{X}$ is given by

$$h(x) := -\log_2 p(x).$$

Definition 2.2 (Shannon Entropy)**.** The Shannon entropy of X is given by the expected information content of an observation of X:

$$H(X) := -\mathbb{E}[h(x)] = -\sum_{x \in \mathcal{X}} p(x) \log_2 p(x).$$

Remark. From now onwards, we refer to the Shannon entropy of a random variable simply as its entropy, and we use base 2 for our logarithms unless otherwise specified.

Remark. Note that the entropy of a random variable is only defined if $p(x) > 0$ for all $x \in \mathcal{X}$.

By Definition 2.1, we see that Propositions 1.1 and 1.2 are satisfied - for example as $p(x)$ decreases, $-\log p(x)$ increases. By Definition 2.2, we see that Propositions 1.3 and 1.4 are, too.

Lemma 2.1 (Discrete Entropy is Non-Negative)**.** *For any discrete random variable X, $h(x) \geq 0$ for all $x \in \mathcal{X}$ and $H(X) \geq 0$.*

Proof. Under the assumption that the probability mass function is nowhere zero, $p(x) \in (0, 1]$ for all $x \in \mathcal{X}$. Therefore $-\log p(x) \geq 0$ and so the information content is non-negative. The entropy is then the expectation over a set of non-negative value. □

We can also validate Proposition 1.5, that the entropy goes to zero if and only if a random variable becomes deterministic. If not deterministic, then this follows immediately:

$$h(x) = -\log p(x) > 0 \text{ for some } x \in \mathcal{X}$$

since $p(x) \in (0, 1)$ for some x for a non-deterministic random variable. We know that the information content is non-negative for all $x \in \mathcal{X}$ and so the expectation over the set must be positive.

To prove the converse:

Lemma 2.2. *Consider a sequence of random variables (X_n) defined on $\mathcal{X} = \{x_1, \ldots, x_m\}$ and with corresponding mass functions p_n, satisfying $p_n(x_1) \to 1$ as $n \to \infty$.*

Then $H(X_n) \to 0$.

Proof. Since we write the entropy as a finite sum, as in Definition 2.2, it suffices to show that each term individually vanishes in the limit.

As $p \to 1$, $p \log p \to 0$, certainly. So the information content of x_1 vanishes.

Now consider $x \in \mathcal{X} \setminus \{x_1\}$. As $n \to \infty$, $p_n(x) \to 0$ for all such x. We apply L'Hôpital's Rule for $\log p$ and $1/p$ to get that

$$\lim_{p \to 0} p \log p = \lim_{p \to 0} \frac{1/p}{-1/p^2} = \lim_{p \to 0} -p = 0.$$

$\square$

Now let us expand the concept of entropy to cases in which we are concerned with multiple random variables. The following definitions extend to up to a finite collection of random variables, however for simplicity we write them for a pair of random variables only.

Consider a pair of discrete random variables X, Y taking values in $\mathcal{X} \times \mathcal{Y}$ and with joint distribution $p(x, y)$.

Definition 2.3 (Joint Entropy)**.** The joint entropy of (X, Y) is given by

$$H(X, Y) := \mathbb{E}[\log p(x, y)] = -\sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x, y) \log p(x, y).$$

Definition 2.4 (Conditional Entropy)**.** The entropy of a random variable X conditioned on another random variable Y is given by

$$\begin{aligned} H(X|Y) &:= \mathbb{E}_Y[\log p(x|y)] \\ &= -\sum_{y \in \mathcal{Y}} \sum_{x \in \mathcal{X}} p(x, y) \log p(x|y), \\ &= -\sum_{y \in \mathcal{Y}} p(y) \sum_{x \in \mathcal{X}} p(x|y) \log p(x|y), \\ &= \sum_{y \in \mathcal{Y}} p(y) H(X|Y = y). \end{aligned}$$

Note that the conditional entropy is a value, not a random variable: we are taking the expectation of a function of the conditional distribution, but not taking a conditional expectation, which would be a function of Y.

Note also that the conditional entropy of X given the event $Y = y$ is a univariate distribution of X, and so the definition follows from the initial definition of Shannon entropy:

Definition 2.5 (Conditional Entropy Given an Event)**.** The entropy of a random variable X conditioned on the event $Y = y$ is given by

$$H(X|Y = y) = E_{p(x|Y=y)}[\log p(x|Y = y)].$$

The chain rule for entropy allows us to decompose joint entropies into a sum of conditional entropies:

Lemma 2.3 (Chain Rule for Entropy [5])**.** *Let us denote the tuple $(X_i, X_{i+1}, \ldots, X_j)$ by (X_i^j). Suppose (X_1^n) is a collection of random variables taking values in $\mathcal{X}_1 \times \cdots \times \mathcal{X}_n$ and with joint distribution $p(x_1, \ldots, x_n)$. Then*

$$H(X_1^n) = \sum_{i=1}^{n} H(X_i|X_1^{i-1}).$$

Proof. We show this inductively. First, for two random variables Y_1, Y_2, we have:

$$\begin{aligned} H(Y_1, Y_2) &= -\mathbb{E}[\log p(y_1, y_2)] \\ &= -\mathbb{E}[\log(p(y_1)) + \log p(y_2|y_1)] \\ &= -\mathbb{E}[\log(p(y_1)] - \mathbb{E}[\log p(y_2|y_1)] \\ &= H(Y_1) + H(Y_2|Y_1). \end{aligned}$$

So this result holds in the base case. Now assume the that chain rule holds for the collection of random variables X_1^{n-1}. Then setting $Y_1 = X_1^{n-1}$ and $Y_2 = X_n$, the same argument as above proves the induction step. □

Corollary 2.4 (Joint Entropy is Linear for Independent Random Variables)**.** *For discrete random variables X, Y, $X \perp Y$,*

$$H(X, Y) = H(X) + H(Y).$$

Proof. By Lemma 2.3, we have that

$$H(X, Y) = H(X) + H(Y|X).$$

By Bayes' Theorem, $p(Y|X) = p(Y)$ if Y and X are independent. Therefore

$$\begin{aligned} H(Y|X) &= -\sum_{x\in\mathcal{X}} p(x) \sum_{y\in\mathcal{Y}} p(y|x) \log p(y|x) \\ &= -\sum_{x\in\mathcal{X}} p(x) \sum_{y\in\mathcal{Y}} p(y) \log p(y) \\ &= -H(Y) \sum_{x\in\mathcal{X}} p(x) = H(Y). \end{aligned}$$

We have shown that $H(Y|X) = H(Y)$: the information of a random variable given an independent random variable equals its unconditional information. □

Remark. This immediately extends to general sequences of independent random variables by decomposition as in the inductive step above. In particular, we get that for (X_1^n) jointly independent,

$$H(X_1^n) = \sum_{i=1}^{n} H(X_i).$$

2.1.1 Differential Entropy and Differences in the Continuous Setting

We can define a concept of entropy in the continuous setting, but the natural extension lacks the properties we would expect for a measure of information. In particular, unlike in the discrete case, differential entropy can be negative since probability density take values greater than 1. However other functions of information do extend to the continuous setting, and admit an expression in terms of this differential form of entropy. We therefore briefly define it here and give an example of a random variable with negative differential entropy.

Definition 2.6 (Differential Entropy)**.** For a continuous random variable X with density function $p(X)$ over a space $\mathcal{X}$, we define the differential entropy of X to be

$$h(X) = -\mathbb{E}[\ln p(x)] = -\int_{\mathcal{X}} p(x) \ln p(x) \ dx.$$

(In the continuous setting, we use the natural log instead of base 2.)

Remark. There is an overlap of notation here with the information content. The difference in capitalised or uncapitalised X does imply the correct formulation, though, and in our discussion the intended use will always be contextually clear.

Example 2.1 (Random Variable with Negative Differential Entropy)**.** Let X be a uniformly-distributed random variable over $[0, 1/2]$. Then

$$h(x) = -\int_0^{1/2} 2\log 2 \ dx = -\log 2.$$

More generally, we see that differential entropy is negative for any uniformly-distributed random variable over an interval of size smaller than one, since the probability density is everywhere greater than one.

2.2 KL Divergence and Mutual Information

So far, we have characterised the information contained in different distributions. We now introduce two functions that allow for comparison between distributions. These are the Kullback-Leibler Divergence, also sometimes called the relative entropy, and the mutual information. Both of these functions, as well as the properties which we shall outline in the remainder of this section, apply to continuous as well as discrete random variables.

Definition 2.7 (Kullback-Leibler Divergence / Relative Entropy)**.** Consider two random variables taking values in the same set $\mathcal{X}$ with corresponding distributions $p(X), q(X)$. Then the KL Divergence between them is given by

$$\mathrm{D_{KL}}(P||Q) := \mathbb{E}_p \left[\log \frac{p(x)}{q(x)} \right],$$

where the subscript reflects that we take the expectation with respect to the distribution $p(X)$.

Remark. The KL divergence does not admit a nice entropy-based definition, since we do not have an expression for $-\mathbb{E}_p[\log q]$. However, in the next chapter, we shall see that it has a natural interpretation in terms of encoding observations as if they were drawn from distribution q when it in fact they were generated by distribution p (Section 3.2.2).

This perspective complements another interpretation of the KL divergence as a measure of the distance between distributions (Lemma 2.10). Importantly, though, note that it is not a metric: it is asymmetric, and the triangle inequality does not hold.

We now define the mutual information. We can view this as a measure of the dependence between random variables (Corollary 2.14).

Definition 2.8 (Mutual Information)**.** The mutual information between two random variables X and Y defined over the same sample space is the divergence between their joint distribution and the product of their marginal distributions:

$$\mathbb{I}(X;Y) = \mathrm{D}_{\mathrm{KL}}(\ p(X,Y) || \ p(X)p(Y)).$$

Example 2.2 (Mutual Information vs. Correlation)**.** A more common measure of dependence between random variables is their Pearson product moment correlation coefficient. Correlation is a measure of linear dependence only, whereas mutual information is a more general measure of dependence [3].

We can also express mutual information in terms of entropy.

Lemma 2.5. *The following are all equivalent to* $\mathbb{I}(X;Y)$:

$$H(X) - H(X|Y),\ H(Y) - H(Y|X),\ H(X) + H(Y) - H(X,Y).$$

Proof. We begin by showing equivalence to $H(X) - H(X|Y)$.

$$\begin{aligned}
\mathbb{I}(X;Y) &= \mathbb{E}\left[\log \frac{p(x,y)}{p(x)p(y)}\right] \\
&= \mathbb{E}\left[\log \frac{p(x|y)}{p(x)}\right] \\
&= -\mathbb{E}[\log p(x)] + \mathbb{E}[\log p(x|y)] \\
&= H(X) - H(X|Y),
\end{aligned}$$

where the final step follows as in the final step of Lemma 2.3.

The exact same argument gives $\mathbb{I}(X;Y) = H(Y) - H(X|Y)$ by flipping the conditioning in the second step above. The final result follows by the chain rule for entropy (Lemma 2.3):

$$\begin{aligned}
H(X) - H(X|Y) &= H(X) + H(Y) - H(Y) - H(X|Y) \\
&= H(X) + H(Y) - H(X,Y).
\end{aligned}$$

Expressing it in this forms allows us to easily see that, unlike divergence, mutual information is symmetric: $H(X) - H(X|Y) = H(Y) - H(Y|X)$ shows that $\mathbb{I}(X;Y) = \mathbb{I}(Y;X)$.

So mutual information is both a measure of the average reduction in our uncertainty about X that comes from knowing Y, and the average reduction in our uncertainty about Y that comes from knowing X. □

A final useful identity for the mutual information comes in the form of another relation between mutual information and divergence:

Lemma 2.6. *For random variables X and Y, let*

$$\mathrm{D}_{\mathrm{KL}}\left(p(Y|x)||\ p(Y)\right)$$

denote the divergence between $p(Y)$ and $p(Y|X=x)$. Let

$$\mathrm{D}_{\mathrm{KL}}\left(p(Y|X)||\ p(Y)\right)$$

denote this divergence as a function of X.

Then

$$\mathbb{I}(X;Y) = \mathbb{E}\left[\mathrm{D}_{\mathrm{KL}}\left(p(Y|X)||\ p(Y)\right)\right].$$

Proof.

$$\begin{aligned}
\mathbb{E}\left[\mathrm{D}_{\mathrm{KL}}\left(p(Y|X)||\ p(Y)\right)\right] &= \int p(x) \int p(y|x) \log \frac{p(y|x)}{p(y)}\ dy\ dx \\
&= \int\int p(x,y) \log \frac{p(x,y)}{p(x)p(y)}\ dy\ dx \\
&= \mathbb{I}(X;Y).
\end{aligned}$$

Note that this naturally aligns with our conception of mutual information as the expected reduction in uncertainty about Y that comes from knowing X. □

Remark. We write this in terms of integrals as a reminder that the results in this section all hold equivalently for continuous random variables. The proof follows identically in the discrete case.

2.2.1 Gibbs' Inequality

We now work towards showing that the KL divergence is a non-negative function. In turn, this will allow us to observe some interesting properties of entropy and mutual information.

First, recall the definition of a convex function:

Definition 2.9 (Convex Function)**.** A function $f : \mathbb{R}^k \to \mathbb{R}$ is convex if for all $x, y \in \mathbb{R}^k$,

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$$

for all $\lambda \in [0,1]$.

Definition 2.10 (Strictly Convex)**.** If the above inequality is strict for all $\lambda \in (0,1)$ and $x \neq y$, then f is said to be strictly convex.

Example 2.3. The function $f(x) = x$ is convex on the reals, as is $f(x) = x^2$. The function $g(x) = \log(x)$ on $(0,\infty)$ is concave in contrast: the inequality goes the other way.

Strict convexity is satisfied by $f(x) = x^2$, but not by $f(x) = x$: here the inequality is in fact everywhere equality. Strict concavity is satisfied by $\log(x)$.

We can view convexity as the property that the line segment connecting x and y lies above or on the graph and concavity as the property that the line segment lies on or below it. Therefore $f(x) = x$ is also concave, $-x^2$ is strictly concave and $-\log(x)$ is strictly convex.

Functions can be neither concave nor convex: $\sin(x)$ is one such example.

The strict convexity of $-\log(x)$ plays an important role in the nature of information.

First, to prove the non-negativity of divergence we begin by showing that for a convex function, the expectation of the image of a random variable is bounded below by the image of its expected value:

Lemma 2.7 (Jensen's Inequality [8])**.** *For a convex function f and a random variable X taking values over some interval I, if the expectation exists* ($\mathbb{E}\left[|X|\right] < \infty$) *and the image of the expectation is finite* ($f\big(\mathbb{E}[|X|]\big) < \infty$)*, then*

$$\mathbb{E}[f(X)] \geq f\big(\mathbb{E}[X]\big).$$

Proof. We will assume without proving the fact that for a convex f, there exists some function $a : I \to \mathbb{R}$ called the sub-derivative of f such that

$$f(x) \geq f(y) + a(y)(x-y) \;\forall\; x \in I$$

provided $f(y) < \infty$.

We also assume without proof that $E[X] \in I$. Then taking x to be an observation of our random variable X and $y = E[X]$, by the existence of the sub-derivative and since $f\big(\mathbb{E}[X]\big) < \infty$ by assumption, we have

$$f(x) \geq f\big(\mathbb{E}[X]\big) + a\big(\mathbb{E}[X]\big)(x - \mathbb{E}[X]) \;\forall x \in I \qquad (*)$$

We look to take expectations on both sides. Then if the expectation on the left hand side exists, we have

$$\mathbb{E}[f(X)] \geq \mathbb{E}\left[f\big(\mathbb{E}[X]\big)\right] + a\big(\mathbb{E}[X]\big)\big(\mathbb{E}[X] - \mathbb{E}[X]\big) = f\big(\mathbb{E}[X]\big)$$

as desired.

Recall that the expectation of $f(X)$ exists if and only if at least one of $\mathbb{E}[f(X)^+]$, $\mathbb{E}[f(X)^-]$ is finite (Section A.2.2). Writing the right hand side altogether as a random variable $Z(X)$, we have $f(X)^- \leq |Z(X)|$ by $(*)$, so it suffices to show that $\mathbb{E}\left[|Z(X)|\right] < \infty$.

$$\mathbb{E}\left[|Z(X)|\right] \leq \left|f\big(\mathbb{E}[X]\big)\right| + \left|a\big(\mathbb{E}[X]\big)\right| \;\big(\left|\mathbb{E}[X]\right| + \mathbb{E}\left[|X|\right]\big),$$

and by our assumptions that $E|X| < \infty$, $f(\mathbb{E}[|X|]) < \infty$, we see that all terms are finite. Therefore $Ef(X)^- \leq E|Z(X)| < \infty$, so the expectation exists and the desired inequality holds. □

With this in hand, we can now prove that divergence is non-negative, and has a unique minimum as a function of one of the distributions:

Theorem 2.8 (Gibbs' Inequality)**.** *For distributions p, q,*

1. $\mathrm{D_{KL}}(p||q) \geq 0$.
2. $\mathrm{D_{KL}} = 0 \iff p = q$.

Proof. We use the fact that $f : \mathbb{R}^+ \to \mathbb{R}$ given by $f(x) = -\log(x)$ is a strictly convex function in order to apply Jensen's Inequality :

$$\begin{aligned}
\mathrm{D_{KL}}(p||q) &= \mathbb{E}_p\left[\log\frac{p(x)}{q(x)}\right] \\
&= \mathbb{E}_p\left[-\log\frac{q(x)}{p(x)}\right] \\
&\geq -\log\mathbb{E}_p\left[\frac{q(x)}{p(x)}\right] \\
&= -\log\sum_{x\in\mathcal{X}} p(x)\frac{q(x)}{p(x)} \text{ or } -\log\left(\int_{\mathcal{X}} p(x)\log\frac{q(x)}{p(x)}\,dx\right) \\
&= -\log 1 = 0.
\end{aligned}$$

If $p = q$, we have $\mathrm{D_{KL}}(p||p) = \mathbb{E}_p[\log 1] = 0$. It remains to prove the the divergence is zero only if the two distributions are identical. We use the strict convexity of $-\log x$ to do so. However we need to assume another property of the sub-derivative: for strictly convex functions, the sub-derivative inequality ($(*)$ in Lemma 2.7) is strict [21].

Consider the proof of Jensen's Inequality (Lemma 2.7) for a strictly convex function. Following back along it, we see that we get the strict inequality

$$f(x) > f\big(\mathbb{E}[X]\big) + a\big(\mathbb{E}[X]\big)(x - \mathbb{E}[X])$$

for all $x \neq \mathbb{E}[X]$. The expectations of both sides of the inequality can therefore only be equal if $X = \mathbb{E}[X]$ with probability 1: in other words if X is constant.

Moving to the proof of the non-negativity of divergence above, we see that

$$\mathrm{D_{KL}}(p||q) = 0 \iff \mathbb{E}_p\left[-\log\frac{q(x)}{p(x)}\right] = -\log\mathbb{E}_p\left[\frac{q(x)}{p(x)}\right].$$

Since $-\log(x)$ is a strictly convex function, this requires that $\frac{q(x)}{p(x)}$ is constant. So $\mathrm{D_{KL}}(p||q) = 0$ if and only if p and q are identical. $\square$

Corollary 2.9. *The mutual information between random variables X and Y is non-negative, 0 if and only if X and Y are independent.*

2.2.2 Convexity of Divergence & Mutual Information

We now show that the KL divergence is jointly convex over the space of pairs of probability distributions. This will help us see how divergence acts as a measure of distance between distributions.

Definition 2.11 (Jointly Convex [5])**.** A bivariate function $f(x, y)$ is said to be jointly convex if for all $\lambda \in [0, 1]$,

$$f\Big(\lambda x_1 + (1-\lambda)x_2, \lambda y_1 + (1-\lambda)y_2\Big) \leq \lambda f(x_1, y_1) + (1-\lambda)f(x_2, y_2).$$

Consider two distributions p and q defined on the same space. By the joint convexity of divergence, any distribution q' formed from a mixture of p and q has lower divergence with p than q does:

Lemma 2.10. *Suppose that the KL divergence is jointly convex. Then for distributions p, q and $q' = \lambda p + (1-\lambda)q$, $\lambda \in [0,1]$,*

$$\mathrm{D}_{\mathrm{KL}}(p||q') \leq \mathrm{D}_{\mathrm{KL}}(p||q).$$

Proof. We have

$$\begin{aligned}
\mathrm{D}_{\mathrm{KL}}(p||q') &= \mathrm{D}_{\mathrm{KL}}(p||\lambda p + (1-\lambda)q) \\
&\leq (1-\lambda)\,\mathrm{D}_{\mathrm{KL}}(p||q) \\
&\leq \mathrm{D}_{\mathrm{KL}}(p||q).
\end{aligned}$$

In particular as $\lambda \to 1$ and the mixture of distributions gets 'closer' to p, the upper bound on the divergence between p and q' goes to 0. □

Remark: We are using joint convexity here over the space of distributions.

To prove the joint convexity of the divergence, we first require the following:

Lemma 2.11 (Log Sum Inequality [24])**.** *For $a_1, \ldots, a_n$, $b_1, \ldots, b_n$ all non-negative real numbers, let*

$$a = \sum_{i=1}^{n} a_i, \quad b = \sum_{i=1}^{n} b_i.$$

Then

$$\sum_{i=1}^{n} a_i \log \frac{a_i}{b_i} \geq a \log \frac{a}{b}.$$

Proof. First, note that $f(x) := x\log(x)$ on $\mathbb{R}^+$ is a convex function: a strictly positive second derivative implies convexity. For the natural log we have $f'' = \frac{1}{x} > 0$ and for all other logs, this is simply scaled by a non-zero constant.

We use a slight variant of Jensen's inequality. Recall that we assumed the existence for any given convex function of a corresponding sub-derivative $a(y)$ such that for all $x \in I$ and for all $y \in I$,

$$f(x) \geq f(y) + a(y)(x-y).$$

Now let $\lambda_1, \ldots, \lambda_n$ be a set of positive constants summing to 1. For a set of points $x_1, \ldots, x_n$ within the range of the convex function f, let $y = \sum_{i=1}^{n} \lambda_i x_i$. Then by repeatedly applying the above inequality and scaling, we get that

$$\sum_{i=1}^{n} \lambda_i f(x_i) \geq f\Bigg(\sum_{i=1}^{n} \lambda_i x_i\Bigg).$$

Now turning to the log sum inequality, have

$$\begin{aligned}
\sum_{i=1}^{n} a_i \log \frac{a_i}{b_i} &= \sum_{i=1}^{n} b_i\Big(\frac{a_i}{b_i} \log \frac{a_i}{b_i}\Big) \\
&= b \sum_{i=1}^{n} \frac{b_i}{b} f\Big(\frac{a_i}{b_i}\Big).
\end{aligned}$$

We take the fractions $\frac{b_i}{b}$ as our constants. Noting that they are all positive and that they sum to 1, we use this variant of Jensen's inequality to get that

$$
\begin{aligned}
b\Bigg(\sum_{i=1}^{n}\frac{b_i}{b}f\big(\frac{a_i}{b_i}\big)\Bigg) &\geq bf\Bigg(\sum_{i=1}^{n}\frac{b_i}{b}\frac{a_i}{b_i}\Bigg)\\
&= bf\Bigg(\sum_{i=1}^{n}\frac{a_i}{b}\Bigg)\\
&= bf\bigg(\frac{a}{b}\bigg)\\
&= a\log\frac{a}{b}.
\end{aligned}
$$

□

With this in hand, joint convexity.

Theorem 2.12 (D_{KL} is Jointly Convex [24]). *For any pair of distribution* (p_1, q_1), (p_2, q_2) *and for all* $\lambda \in [0, 1]$,

$$
\mathrm{D}_{\mathrm{KL}}\Bigg(\lambda p_1 + (1-\lambda)p_2 \;||\; \lambda q_1 + (1-\lambda)q_2\Bigg) \;\leq \lambda\,\mathrm{D}_{\mathrm{KL}}(p_1||q_1) + (1-\lambda)\,\mathrm{D}_{\mathrm{KL}}(p_2||q_2).
$$

Proof. We apply the log sum inequality. Let p_m, q_m denote the mixture of the distributions (p_1, p_2), (q_1, q_2), respectively.

$$
\begin{aligned}
\mathrm{D}_{\mathrm{KL}}(p_m||q_m) &= \mathbb{E}_{p_m}\left[\log\frac{p_m}{q_m}\right]\\
&\leq \lambda\,\mathbb{E}_{p_1}\big[\frac{p_1}{q_1}\big] + (1-\lambda)\,\mathbb{E}_{p_2}\big[\frac{p_2}{q_2}\big]\\
&= \lambda\,\mathrm{D}_{\mathrm{KL}}(p_1||q_1) + (1-\lambda)\,\mathrm{D}_{\mathrm{KL}}(p_2||q_2).
\end{aligned}
$$

□

Although we would like to also use this joint convexity argument as justification for why mutual information is a measure of the dependence of random variables, it does not transfer immediately. The form of the corresponding result is slightly different.

Lemma 2.13 (Convexity of Mutual Information [1]). *Suppose* X, Y *are two random variables with joint distribution* $p(X)p(Y|X)$*. Then for a fixed marginal distribution of* X, $\mathbb{I}(X;Y)$ *is a convex function of* $p(Y|X)$.

Proof. Fix $p(X)$ and consider conditional distributions $p_1(Y|X), p_2(Y|X)$. Consider for $\lambda \in [0, 1]$ the mixture distribution

$$
p_\lambda(Y|X) = \lambda p_1(Y|X) + (1-\lambda)p_2(Y|X).
$$

We first aim to show that both the joint mixture distribution and the marginal mixture distributions are linear functions of p_1 and p_2. For the joint distribution, we have

$$
p_\lambda(X, Y) = \lambda p_1(X, Y) + (1-\lambda)p_2(X, Y)
$$

immediately. Integrating the joint distribution with respect to X gives us the marginal $p_\lambda(Y)$ and by linearity of integration, we can conclude that $p_\lambda(y)$ is linear in p_1 and p_2. We can write

$$q_\lambda(X,Y) = p(X)p_\lambda(Y) = \lambda q_1(X,Y) + (1-\lambda)q_2(X,Y).$$

Therefore

$$\begin{aligned}\mathbb{I}(X;Y_\lambda) &= D_{KL}\big(p_\lambda(X,Y)||\ q_\lambda(X,Y)\big)\\ &= D_{KL}\big(\lambda p_1(X,Y) + (1-\lambda)p_2(X,Y)||\ \lambda q_1(X,Y) + (1-\lambda)q_2(X,Y)\big).\end{aligned}$$

By Lemma 2.12, we therefore have that

$$\mathbb{I}_\lambda(X;Y_\lambda) \leq \lambda\, \mathbb{I}_{p_1}(X;Y) + (1-\lambda)I_{p_2}(X;Y).$$

where $\mathbb{I}_{p_1}, \mathbb{I}_{p_2}$ denote that we are considering $Y|X \sim p_1(Y|X)$ and $\sim p_2(Y|X)$ respectively. ☐

This allows us to see mutual independence as a measure of dependence between distributions. Our desired interpretation follows from considering the set of minima, given by the space of functions independent to X, instead. For example, we have the following:

Lemma 2.14. *Suppose there exists random variables X, Y and Z, $X \perp Z$. For a random variable $Y' = \lambda Y + (1-\lambda)Z$, $\lambda \in [0,1]$,*

$$\mathbb{I}(X;Y') \leq \mathbb{I}(X;Y)$$

Proof. Similar to above, this follows immediately by applying Lemma 2.13 and recalling Corollary 2.9. ☐

The random variable Y' has mixture distribution $\lambda p(Y) + (1-\lambda)p(Z)$. In taking Y to Y', the dependence on X is reduced by the variability independent of X introduced by Z. As the mixture gets 'closer' to being independent of X, the mutual information decreases.

2.2.3 Conditional Divergence and Information

We now extend the definitions of divergence and mutual information to deal with conditional distributions.

Definition 2.12 (Conditional Divergence)**.** For distributions $p(x,z)$, $q(x,z)$, the conditional divergence between $p(x|z)$ and $q(x|z)$ is given by

$$\mathrm{D_{KL}}\left(p(x|z)\ ||\ q(x|z)\right) := E_{p(x,z)} \log \frac{p(X|Z)}{q(X|Z)}.$$

Definition 2.13 (Conditional Mutual Information)**.** The conditional mutual information between random variables X and Y conditioned on a random variable Z is given by

$$\begin{aligned}\mathbb{I}(X;Y|Z) &:= \mathbb{E}_{p(x,y,z)} \log \frac{p(X,Y|Z)}{p(X|Z)p(Y|Z)} \\ &= \mathrm{D}_{\mathrm{KL}}\left(p(X,Y|Z) \mid\mid p(X|Z)p(Y|Z)\right) \\ &= H(X|Z) - H(X|Y,Z).\end{aligned}$$

To reframe in terms of our previous descriptions, we can see the conditional mutual information as the reduction in uncertainty about X upon observing Y when a random variable Z is known prior. Equivalently, it is the level of the dependence of X on Y given that Z is already known.

Corollary 2.15. *Conditional mutual information is non-negative, 0 if and only if X and Y are conditionally independent.*

These definitions allow us to define a chain rule for mutual information. Similar to entropy, this proves useful for decomposing the dependence between a number of random variables.

A nice interpretation of this is as a proof of equivalence between the information gain about one variable upon observing a number of other random variables simultaneously and the information gain about the same variable upon observing the same set of random variables sequentially, given that the variables are iid.

Lemma 2.16 (Chain Rule for Mutual Information [5])**.** *For random variables (X_1^n, Y), the mutual information between the set (X_1^n) and Y is given by*

$$\mathbb{I}(X_1^n;Y) = \sum_{i=1}^{n} \mathbb{I}(X_i;Y|X_1^{i-1}).$$

Proof. We use the chain rule for entropy (Lemma 2.3) and the entropy formulae for mutual and conditional mutual information (Lemma 2.5):

$$\begin{aligned}\mathbb{I}(X_1^n;Y) &= H(X_1^n) - H(X_1^n|Y) \\ &= \sum_{i=1}^{n} H(X_i|X_1^{i-1}) - \sum_{1}^{n} H(X_i|X_1^{i-1},Y) \\ &= \sum_{i=1}^{n} \mathbb{I}(X_i,Y|X_1^{i-1}).\end{aligned}$$

□

Note that equivalently, by the symmetry of mutual information,

$$\mathbb{I}(X;Y_1^n) = \sum_{i=1}^{n} \mathbb{I}(X;Y_i \mid Y_1^{i-1}).$$

2.3 Returning to Entropy

With these characterisations of divergence and information in hand, we confirm a pair of intuition-motivated properties about discrete entropy. We first formalise Proposition 1.6 by proving that the expected information content of a discrete uniform random variable is the maximum for any distribution over that sample space. We then show that conditioning on the information in another random variable reduces the expected information content.

Lemma 2.17 (Uniform Random Variables Maximise Entropy [5])**.** *For a discrete random variable taking values in $\mathcal{X}$, $H(X) \leq \log|\mathcal{X}|$, with equality if and only if X is uniformly distributed.*

Proof. Consider an arbitrary distribution $p(X)$ over $\mathcal{X}$ and let $u(X)$ be the uniform distribution over the sample space: $u(x) = \frac{1}{|\mathcal{X}|}$ for all $x \in \mathcal{X}$. Consider the divergence between p and u and by Gibbs' Inequality (Theorem 2.8):

$$
\begin{aligned}
\mathrm{D}_{\mathrm{KL}}(p||u) &= \sum_{\mathcal{X}} p(x) \log \frac{p(x)}{u(x)} \\
&= \sum_{\mathcal{X}} p(x) \log p(x) + \sum_{\mathcal{X}} p(x) \log|\mathcal{X}| \\
&= -H(X) + \log|\mathcal{X}| \geq 0.
\end{aligned}
$$

□

This also shows without needing any calculation that $H(X) = \log|\mathcal{X}|$ for a uniform random variable since by Theorem 2.8, $\mathrm{D}_{\mathrm{KL}}(p||u) = 0$ when $p = u$.

Remark. One interesting contrast between discrete and differential entropy is that differential entropy is maximised by the normal distribution, instead of the uniform distribution [5]. We omit the proof.

Lemma 2.18 (Conditioning Reduces Entropy [5])**.** *Conditioning reduces entropy: For a discrete random variable X, for all random variables Y,*

$$H(X) \geq H(X|Y).$$

The two are equal if and only if X is independent of Y.

Proof. $H(X) - H(X|Y) = \mathbb{I}(X;Y) \geq 0$, 0 if and only if X and Y are independent by Corollary 2.9. □

Corollary 2.19. *Conditioning reduces differential entropy.*

Proof. As we noted, the identities for mutual information hold for both discrete and differential entropy. Therefore the result follows exactly as above. □

2.4 The Data-Processing Inequality

We now begin to turn towards the study of how information can be effectively represented, which is the domain of coding and compression. We complete this chapter by considering the limits on the information exchanged between successive random variables.

In particular, suppose that we are interested in making inference about some random variable X, but can only observe some related variant Y. The data-processing inequality states that no transformation of Y can fundamentally increase the information it contains about X. Any additional operations can only lose information about it.

To state the inequality precisely:

Theorem 2.20 (Data-Processing Inequality [19])**.** *Let X, Y, Z be random variables such that $X \perp Z|Y$. Then*

$$\mathbb{I}(X;Z) \leq \mathbb{I}(X;Y)$$

and

$$\mathbb{I}(X;Z) \leq \mathbb{I}(Y;Z).$$

Proof. By the chain rule for mutual information, we have

$$\begin{aligned}\mathbb{I}(X;Z) \leq \mathbb{I}(X;Z) + I(X;Y|Z) &= \mathbb{I}(X;Y,Z)\\ &= \mathbb{I}(X;Y) + \mathbb{I}(X;Z|Y)\\ &= \mathbb{I}(X;Y),\end{aligned}$$

where the final step follows from Lemma 2.5, the conditional independence assumption, and the equality of conditional entropy in Lemma 2.18:

$$I(X;Z|Y) = H(X|Y) - H(X|Y,Z) = H(X|Y).$$

Similarly,

$$\begin{aligned}\mathbb{I}(X;Z) \leq \mathbb{I}(X;Z) + I(Y;Z|X) &= \mathbb{I}(X,Y;Z)\\ &= \mathbb{I}(Y;Z) + \mathbb{I}(X;Z|Y)\\ &= \mathbb{I}(Y;Z).\end{aligned}$$

□

Remark. The second inequality states that the information between X and Z can be at most the information Y shares with Z. This follows since Z receives no other information about X.

Example 2.4. A useful analogy here is to the film trope of someone starting with a blurry image, but then zooming in to a section of the picture and managing to view a high resolution version of this section. The data processing inequality proves that this is impossible.

Here X would be a clear version of this section, Y the blurry image and Z our zoomed in version of it. If zooming in created a clearer version of the picture, then the mutual information between X and Z would be greater than that between X and Y.

2.4.1 Summary

Within this section, we have explored the theory of information that arises from Shannon's definition of entropy. We have validated that Shannon entropy satisfies the intuitive properties outlined in Section 1.1, and shown how the associated properties of divergence and mutual information can act as measures of distance between distributions and independence between random variables, respectively.

We have now begun to consider how information can be exchanged. While the Data-Processing Inequality (Theorem 2.20) describes how information flows in a system with multiple random variables, we shall begin our discussion of compression by considering how to efficiently encode data generated by a single variable. We shall see that the difficulty in describing each outcome, as quantified by the length needed to encode it, is fundamentally tied to the random variable's entropy - its Shannon entropy, specifically - bringing us back to Proposition 1.7 and the idea of entropy as a measure of complexity.

Chapter 3

Coding & Compression

We shall begin here by considering the limiting behaviour of sequences of information. This will lead us to Shannon's Source Coding Theorem (Theorem 3.2), one of the fundamental results connecting coding and entropy. We turn to practical approaches to lossless compression - compressions for which each given outcome is uniquely described - and again consider the limiting behaviour. Through the Kraft-McMillan Inequalities (Theorems 3.3, 3.4), we shall tighten the connection between compression and prediction, before once again seeing that entropy is fundamental (Theorem 3.5). Finally we provide an example of a near-optimal practical compressor (Lemma 3.9) and very briefly describe some of the challenges of language modelling. The argument here is an adaptation of [12].

3.1 Asymptotic Limits on Information and Compression

3.1.1 The Asymptotic Equipartition Principle

Consider a discrete random variable X with an alphabet of possible values $\mathcal{X} = \{x_1, \ldots, x_n\}$ and distribution $p(X) := \{p_1, \ldots, p_n\}$.

Suppose that we observe a sequence $(X_1, \ldots, X_N)$ of realisations of X, each observation iid. Let us consider the likelihoods of different sequences.

First, imagine what we would expect an 'average' sequence to look like. Within a sequence of N observations, the number we would expect to have value x_i is given by Np_i. We define a sequence with such frequencies to be the 'typical' sequence and denote it x_{typ}.

Now since the random variables in the sequence are i.i.d., the likelihood of a given sequence is the product of the likelihoods of each observation. Therefore the likelihood of a typical sequence is given by

$$P(x_{\text{typ}}) = \Pi_{i=1}^{n} p_i^{Np_i},$$

with corresponding information content

$$h(x_{\text{typ}}) = -N \sum_{i=1}^{n} p_i \log p_i = NH(X).$$

Note that we can therefore more efficiently write the likelihood of a typical sequence as

$$P(x_{\text{typ}}) = 2^{-NH(X)}.$$

The asymptotic equipartition principle states that as N increases, the likelihoods of sequences concentrate around that of the typical sequence with increasingly high probability. Equivalently, as N increases, the distribution of the information content of sequences concentrates around the information content of the typical sequence, and almost all sequences become almost equally informative. The principle can be seen as an analogue to the weak law of large numbers for information.

Theorem 3.1 (Asymptotic Equipartition Principle)**.** *For $X_1, \ldots, X_N$ i.i.d., with distribution $p(X)$,*

$$-\log p(X_1, \ldots, X_n) \to NH(X) \text{ in probability.}$$

Proof. $\log p(X_1), \ldots, \log p(X_N)$ are functions of independent random variables and so are themselves independent random variables. We can therefore apply the weak law of large numbers to this sequence to get

$$\begin{aligned} -\frac{1}{N} \log(p(X_1, \ldots, X_N)) &= -\frac{1}{N} \sum_{i=1}^{N} \log p(X_i) \\ &\to -\mathbb{E}[\log p(X)] = H(X) \text{ in probability.} \end{aligned}$$

□

We can expand our focus from the typical sequence to the set of sequences closest to it, with respect to their likelihoods. We call this the typical set:

Definition 3.1 (Typical Set)**.** The typical set with tolerance β, $T_{N\beta}$, for a sequence $X_1, \ldots, X_n$ is composed of those elements with likelihoods differing by a factor of at most $2^{-N\beta}$ from the likelihood of the typical sequence:

$$T_{N\beta} = \left\{ x \in \mathcal{X}^N : \left| -\frac{1}{n} \log P(X) - H \right| < \beta \right\}.$$

3.1.2 Typicality & Shannon's Source Coding Theorem

Consider how to best encode the outcomes of the sequence $(X_1, \ldots, X_N)$. One way to ensure each possible sequence had a unique encoding would be to assign to each possible sequence - equivalently, to each element of $\mathcal{X}^N$ - an arbitrary, fixed-length bit string. This is called a block coding scheme. The length of the encoding needed is determined by the size of the alphabet.

Definition 3.2 (Raw Bit Content)**.** The raw bit content of a random variable X with alphabet $\mathcal{X}$ is given by

$$H_0(X) := \log |\mathcal{X}|.$$

The raw bit content defines a lower bound on the number of bits needed to assign a fixed-length encoding to each element of $\mathcal{X}$.

Remark. Returning to our discussion in Section 1.1, note that the raw bit content is equivalently a lower bound on the number of binary questions needed to specify each element of $\mathcal{X}$ uniquely.

Our block coding scheme then requires at least $NH_0(X)$ bits, since

$$\log |\mathcal{X}^N| = \log |\mathcal{X}|^N = N \log |\mathcal{X}|.$$

This is a strict lower bound for lossless block encodings. Lossy compressions - encodings where some sequences are mapped to the same string, or simply ignored altogether - can achieve greater efficiency, at the cost of accuracy.

To explore the relation between those two factors, we aim to maximise accuracy within an efficiency constraint. We focus on the smallest subsets of sequences containing a given proportion of the total likelihood. For an encoding C which does not correspond to a unique $x \in \mathcal{X}$ or $x^N \in \mathcal{X}^N$, we say that all strings which are compressed to C are non-decodable, and that all such strings incur an error.

Definition 3.3 (Smallest δ-sufficient subset)**.** The smallest δ-sufficient subset, S_δ, of $\mathcal{X}^N$ for $\delta \in (0,1)$ is the smallest subset of $\mathcal{X}^N$ such that

$$P(x \in S_\delta) \geq 1 - \delta.$$

Definition 3.4 (Essential Bit Content)**.** For an tolerance threshold δ, the essential bit content is given by

$$H_\delta(X) := \log |S_\delta|.$$

The essential bit content therefore provides a lower bound on the cost of block encoding the smallest δ-sufficient subset.

For small N, H_δ varies greatly as δ does over $(0,1)$. However as N grows large, we find that remarkably, the tolerance permitted loses importance. Regardless of the value of δ within $(0,1)$, we find that the limit of the essential bit content is $NH(X)$:

Theorem 3.2 (Shannon's Source Coding Theorem)**.** *Let X be a random variable, taking on values in $\mathcal{X}$ and with entropy $H = H(X)$. For any pair $\varepsilon > 0$ and $\delta \in (0,1)$, there exists $N_0 \in \mathbb{N}$ such that for $N > N_0$,*

$$|\frac{1}{N} H_\delta(X^N) - H| < \varepsilon.$$

Proof. Fix ε, δ. First, we look to show that for sufficiently large N,

$$\frac{1}{N} H_\delta(X^N) < H + \varepsilon.$$

We do so using the typical set.

Let $\beta = \varepsilon$ and consider $T_{N\varepsilon}$. By Theorem 3.1, there exists N_0 such that

$$N > N_0 \implies P(x \in T_{N\varepsilon}) \geq 1 - \delta.$$

Now by definition of S_δ as the smallest subset containing probability mass of at least $1-\delta$, for all $N > N_0$, $|S_\delta| \leq T_{N\varepsilon}$. Therefore calculating the maximum possible size of $T_{N\varepsilon}$ provides an upper bound for S_δ for that N.

The minimum likelihood of a sequence in $T_{N\varepsilon}$ is given by $2^{-N(H+\varepsilon)}$ and the total likelihood of the elements in $T_{N\varepsilon}$ cannot be greater than one. Therefore

$$\begin{aligned} |T_{N\varepsilon}| \leq 2^{N(H+\varepsilon)} &\implies |S_\delta| \leq 2^{N(H+\varepsilon)} \\ &\implies H_\delta(X^N) \leq N(H+\varepsilon). \end{aligned}$$

Now let us turn to the lower bound

$$\frac{1}{N} H_\delta(X^N) > H - \varepsilon.$$

We again argue using the typical set.

Suppose a set S'_δ exists such that $|S'_\delta| \leq 2^{N(H-\varepsilon)}$. We look to provide an upper bound on the probability for sufficiently large N that an arbitrary sequence lies in S'_δ. To this end, we separate S'_δ into two disjoint subsets and calculate the respective probabilities of a sequence lying in each. In particular, we use the typical set for $\beta = \varepsilon/2$ and consider whether sequences in S'_δ lie in $T_{n\varepsilon/2}$ or not:

$$\begin{aligned}
P(x \in S'_\delta) &= P(x \in S'_\delta \cap T_{n\varepsilon/2}) + P(x \in S'_\delta \cap \mathrm{T_{n\varepsilon/2}}^c) \\
&\leq |\,\mathrm{S}'_\delta\,| 2^{-N(H-\varepsilon/2)} + P(x \notin \mathrm{T_{n\varepsilon/2}}) && (a) \\
&\leq 2^{-N\varepsilon/2} + P(x \notin \mathrm{T_{n\varepsilon/2}}) && (b),
\end{aligned}$$

where (a) follows because the probability of an element in $T_{n\varepsilon/2}$ is at most $2^{-N(H-\varepsilon/2)}$ (Definition 3.1) and (b) follows by our assumption on the size of S'_δ.

Certainly $2^{-N\varepsilon/2}$ vanishes in N, and by Theorem 3.1 we know that $P(x \notin T_{n\varepsilon/2})$ does likewise. Therefore for any set with size less than $2^{N(H-\varepsilon)}$, the probability that an arbitrary sequence is decodable goes to 0 in the limit: there exists $N_0 \in \mathbb{N}$ such that for all $N > N_0$, for any set S'_δ with size less than $2^{N(H-\varepsilon)}$, $P(x \in \mathrm{S}'_\delta) < 1 - \delta$. Consequently for all $N > N_0$, $|S_\delta| > 2^{N(H-\varepsilon)}$. The result follows. □

It is worthwhile to stop and admire this for a moment. We have shown here that entropy is in a sense the fundamental limit of lossless compression. Provided our encoding has a per-symbol length any greater than $H(X)$, it can achieve asymptotically vanishing loss. Note in particular that unless the distribution is uniform,

$$H(X) < H_0(X)$$

and therefore, especially for large N,

$$2^{NH(X)} << 2^{NH_0(X)}.$$

We can achieve this enormous improvement despite still using fixed-length block encodings.

Conversely, since Theorem 3.2 holds for all δ-sufficient subsets, no matter how much accuracy you are willing to sacrifice, in the limit you still cannot escape the entropy bound.

3.2 Lossless Compression

3.2.1 Prefix Codes

We have a sharp phase change for block coding with respect to the limits for lossless and lossy compression: for a sequence of i.i.d. samples of a random variable X, if allowed an error with probability $\delta > 0$, the limit on per-symbol code length is the entropy. For $\delta = 0$, however, by simple combinatorial constraints we need a length of $|\,\mathcal{X}\,|^n$.

In this section, we focus on the limits of lossless compression under schemes other than block coding. We shall first consider variable-length symbol coding. Under such a scheme, each member of the alphabet $\mathcal{X}$ is uniquely mapped to a binary string, rather than each n-length block. Encoding by symbol allows us to work with the same coding scheme for different length sequences, so is of significant practical benefit.

We aim to reduce the length of our coding by mapping more probable outcomes shorter strings. Before determining the optimal approach, let us first consider the criteria our code must satisfy. The most important of these is unique decodability: any scheme must be an injective map from sequences to codes. One approach that ensures this is the construction of a prefix code.

Definition 3.5 (Prefix Code)**.** A prefix code is one for which no binary encoding of a symbol is a prefix of the binary encoding for another symbol.

Example 3.1. Suppose we had some alphabet $\mathcal{X}$ taking on values $\{x_1, x_2, \ldots, x_n\}$. Suppose that we constructed a prefix code which mapped x_1 to the string 10. Then none of $\{x_2, \ldots, x_n\}$ could have encodings which began 10, nor have the single-bit encoding 1 (since then they would be a prefix for x_1).

Note that prefix codes are a subset of uniquely decodable codes. Given a string encoded by a prefix code, we can decode it by repeatedly going along the string until we find a sequence which uniquely corresponds to a given symbol. We shall show that the restriction to prefix codes does not reduce the optimal efficiency of our encodings.

3.2.2 The Kraft-McMillan Inequality

Let us expand on the previous example slightly. Consider X taking values in $\mathcal{X} = \{x_1, \ldots, x_n\}$ and with distribution $p(X)$ as usual. We know that we need $|\,\mathcal{X}\,|$ different binary strings in order to uniquely encode every symbol. For a block encoding, this required blocks of size $\log|\,\mathcal{X}\,|$. For a prefix code, we are attempting to do better than this on average. In particular, letting $l(x_i) = l_i$ and $p(x_i) = p_i$, we are attempting to minimise

$$L(C, X) = \sum_{i=1}^{n} l_i p_i$$

over all codewords C.

Continuing towards optimality, we next need to determine the set of code lengths for which a prefix code is possible. This will then provide a constraint under which we can optimise with respect to the distribution of our random variable. The challenge of prefix schemes is that in reducing the length of one symbol, we force the length of others to increase:

Example 3.2. Suppose we have an alphabet $\mathcal{X} = \{x_1, x_2, \ldots, x_{16}\}$. Suppose under our block encoding scheme x_1 had code 1101, x_2 had code 1110, x_3 corresponded to the string 1100 and x_4 to 1111.

Believing x_1 to have higher probability, we want to change it to have code 11. However this forces the length of all other codes to increase in the following manner: x_1 is now a prefix for the current codes of x_2, x_3 and x_4. Therefore we must change their encodings. But they cannot switch to be any other 4-length code, since these all already correspond to symbols. Moreover, if they became 5-length codes, then some other symbols would now be prefixes for them. Compressing x_1 causes the length of the encodings of these other variables to increase.

Finding an optimal prefix code involves identifying when this 'ripple effect' is worthwhile, and which other codes should be affected by it. The bounds on this are provided by the Kraft Inequality:

Theorem 3.3 (Kraft Inequality). *Suppose a prefix code has lengths $(l_1, \ldots, l_n)$ for the encodings of symbols $(x_1, \ldots, x_n)$. Then*

$$\sum_{i=1}^{n} 2^{-l_i} \leq 1.$$

Moreover for a set of integer code lengths satisfying this inequality, there exists a prefix code with those code lengths.

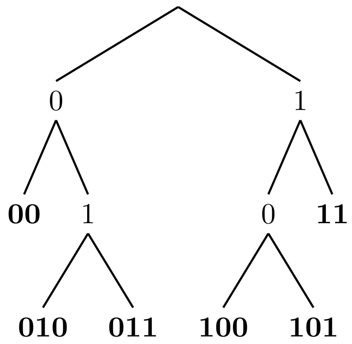

Figure 3.1: A prefix code represented as the leaves of a binary tree. This particular prefix code is given by $\{00, 010, 011, 100, 101 \text{ and } 11.\}$

Proof. We shall visualise our argument here by considering binary trees. A binary tree is a hierarchical structure consisting of nodes, each of which either has two descendants or none. If none, then the node is called a leaf. Each node has a unique parent, except the root node, which has no parent nodes. We can represent the codewords in a prefix code as the leaves on such a tree. The tree corresponding to our prefix code has leaves at the nodes corresponding to each binary string, as in Figure 3.1. Therefore for each codeword of length k, there exists a leaf at depth k.

Consider the encoding for some symbol x_i. In particular, consider the proportion of total possible encodings that this codeword takes up due to the conditions on prefix codes. For example, 00 takes up 1/4 of all possible encodings. Over all nodes at a given depth, the total proportion is 1: if every leaf is a node then the tree must end at that depth. The proportion for a given leaf is therefore the reciprocal of the number of the leaves at a given depth. Hence if at depth l_i, the proportion taken up is 2^{-l_i}.

Now since a prefix code, we know that proportions taken up by each encoding do not overlap. The total proportion taken up is therefore the sum of the individual proportions and certainly can at most equal 1, at which point our prefix code has taken up all possible encodings. Consequently, we get that

$$\sum_{i=1}^{n} 2^{-l_i} \leq 1,$$

as desired.

For a given set of integer lengths $\{l_1, \ldots, l_n\}$ satisfying this inequality, we can assign them to leaves at depths equal to their lengths. This is certainly possible by the proportion argument above: we will not run out of valid leaves at any point.

Consider the encoding for a length l_i. There are no descendants of a leaf, so this encoding is not a prefix for any other. Since this holds for all lengths in our set, we see that this is indeed a prefix code. □

This result in fact characterises not only prefix codes, but also all uniquely decodable codes:

Theorem 3.4 (McMillan)**.** *Suppose a uniquely decodable code $\mathcal{C}$ has lengths $(l_1, \ldots, l_n)$ for the encodings of symbols $\mathcal{X} = \{x_1, \ldots, x_n\}$. Then*

$$\sum_{i=1}^{n} 2^{-l_i} \leq 1.$$

Moreover for a set of integer code lengths satisfying this inequality, there exists a prefix code with those code lengths.

Proof. Let $\mathcal{C}_k$ denote all codes that are generated by length k strings of symbols from $\mathcal{X}^k$. Let the number of sequences with length l codewords be given by n_l and note that since $\mathcal{C}_l$ is uniquely decodable, $n_l \leq 2^l$. Let $l_{\max}$ be the length of the longest code for a single symbol. Summing first over k-length strings, then switching to summing over code lengths, we obtain

$$\begin{aligned}
\left(\sum_{\mathcal{X}} 2^{-l_i}\right)^k &= \sum_{\mathcal{X}^k} 2^{-l(\mathcal{C}_k)} \\
&= \sum_{l} n_l 2^{-l} \\
&\leq \sum_{l} 2^l 2^{-l} \\
&= \sum_{i=1}^{kl_{\max}} 1 \\
&= kl_{\max}.
\end{aligned}$$

This gives us

$$\left(\sum_{\mathcal{X}} 2^{-l_i}\right) \leq (kl_{\max})^{1/k}.$$

Note that the left side does not depend on k, and that

$$\lim_{k\to\infty} (\alpha k)^{1/k} = 1$$

for any $\alpha \in \mathbb{R}^+$. Therefore taking k to ∞, we get the desired inequality.

This shows that the code lengths for any uniquely-decodable code must satisfy the Kraft Inequality. By Theorem 3.3, for any such set of lengths, there exists a corresponding prefix code with exactly the same code lengths. □

The code length condition of Theorem 3.3 is therefore not only necessary for any lossless prefix code, but also sufficient: any set of lengths satisfying the inequality has a lossless prefix code equivalent.

With this in hand, we can finally derive the optimal encoding length for a given distribution. Entropy, once again, acts as a lower bound, this time for the length of any uniquely decodable code.

Theorem 3.5 (Source Coding Theorem for Symbols)**.** *Consider a random variable X, corresponding alphabet $\mathcal{X} = \{x_1, \ldots, x_n\}$ and corresponding distribution $p(X)$, $p_i := p(x_i)$. For every uniquely-decodable code C for X which has lengths $\{l_1, \ldots, l_n\}$, $l_i := l(x_i)$,*

$$H(X) \leq L(C, X).$$

Moreover there exists a code $\mathcal{C}$ such that

$$H(X) \leq L(\mathcal{C}, X) < H(X) + 1.$$

Proof. By Theorem 3.4, we can focus on the set of prefix codes. We consider the lower bound first. We define the implicit probabilities of an encoding to be the proportion of all binary strings that the encoding takes up. As argued in Theorem 3.3, the sum of the proportions is at most 1, so these can indeed represent probabilities.

We denote the implicit probability for a code word of length l_i by q_i. Recall that $q_i = 2^{-l_i}$ (Theorem 3.3). Then for any code C we have

$$\begin{aligned} L(C, X) &= \sum_{i=1}^{n} p_i l_i = \sum_{i=1}^{n} -p_i \log q_i \\ &\geq \sum_{i=1}^{n} -p_i \log p_i && (a) \\ &= H(X), \end{aligned}$$

where (a) follows from Gibbs' Inequality (Theorem 2.8).

To show the upper bound, we consider a code with lengths slightly longer than those implied by the probabilities of the corresponding symbols. This gives us integer codelengths:

$$l_i' := \lceil -\log p_i \rceil.$$

Since each length is either staying the same (if already an integer) or getting longer, this certainly still satisfies the Kraft Inequality:

$$\sum_{i=1}^{n} 2^{-l_i'} \leq \sum_{i=1}^{n} 2^{-l_i} = \sum_{i=1}^{n} p_i = 1,$$

so there exists a prefix code $\mathcal{C}$ with these codelengths by Theorem 3.4.

Moreover the expected length of this is bounded above by $1 + H(X)$:

$$L(C, X) = \sum_{i=1}^{n} p_i \lceil -\log p_i \rceil < \sum_{i=1}^{n} p_i(-\log p_i + 1) = H(X) + 1.$$

□

Corollary 3.6. *For any sequence of length N generated by observations of X and for any lossless encoding C,*

$$H(X^N) \leq L(C, X^N).$$

Proof. Apply Theorem 3.5 to the random variable X^N with alphabet $\mathcal{X}^N$ with distribution $p(x^N)$. Note in particular that entropy appears as a fundamental limit for non-i.id. sequences, too. □

Remark. In Theorem 3.5 and Corollary 3.6, equality is achieved if and only if the implicit probabilities are equal to the true probabilities for every symbol. Equivalently, equality is achieved if and only if the length of each codeword is equal to the information content of each symbol:

$$p_i = q_i = 2^{-l_i} \implies l_i = -\log p_i.$$

This is not always possible. Achieving a length equal to the entropy cannot be done unless it is the case that all probabilities in the distribution are negative powers of 2.

We can now also provide a justification for the framing of KL divergence and mutual information as measures of the cost of coding.

Proposition 3.7. *The KL divergence between distributions p and q is the cost of encoding data generated by p using a coding scheme $\mathcal{C}$ with implicit probabilities q.*

Proof. The length of our prefix code is given by

$$\begin{aligned} L(\mathcal{C}, X) &= \sum_{i=1}^{n} -p_i \log q_i \\ &= \sum_{i=1}^{n} -p_i \log p_i \frac{q_i}{p_i} \\ &= \sum_{i=1}^{n} -p_i \log p_i + \sum_{i=1}^{n} p_i \frac{p_i}{q_i} \\ &= H(X) + \mathrm{D}_{\mathrm{KL}}(p||q), \end{aligned}$$

and by Theorem 3.5, we know that $H(X)$ is the optimal length code, achieved if and only if we encode with implicit probabilities equal to the true distribution p. □

Corollary 3.8. *Mutual information is the cost of encoding according to the product of the marginal distributions instead of the joint distribution.*

In Lemma 2.12, we formalised the notion of divergence as a measure of distance between distributions. Proposition 3.7 then formalises the equivalence between compression and prediction. For any distribution close to the true distribution, there exists an associated coding scheme which encodes almost optimally since the divergence between distributions is low; for any encoding which achieves near-optimal performance, its implicit distribution must have low divergence and thus be close to the true distribution.

3.2.3 Towards Optimal Compression

Theorem 3.5 showed that the entropy limit on uniquely decodable codes is tight. However symbol coding is far from optimal in many scenarios.

Example 3.3. Suppose that we wanted to describe the repeated flips of a coin, which we represent by a random variable X. The set of outcomes is given by {Heads, Tails, Landing on its sid with corresponding probabilities $\{\frac{50}{101}, \frac{50}{101}, \frac{1}{101}\}$. The optimal prefix code for such a random variable would assign to each of these outcomes a length-2 code: say $\{00, 01, 10\}$. The entropy of a coin flip is given by

$$H(X) = -\frac{100}{101} \log \frac{50}{101} - \frac{1}{101} \log \frac{1}{101} \approx 1.07.$$

Suppose that we looked to a compress a sequence of 100 independent realisations of the random variable. Then the entropy of this sequence would be

$$H(X^{100}) = 100\ H(X) \approx 107.$$

(Note the first equality follows by independence).

With our optimal prefix scheme, however, we require 200 bits to transmit the information.

The upper bound of $H(X)+1$ for a prefix code can make a significant difference for encoding large sequences of data, especially when the entropy of the random variable itself is small.

Moreover, prefix codes do not extend well to non-i.i.d. data, since their encodings remain fixed. In such environments, we can improve efficiency significantly by coding according to conditional likelihood instead of relative frequency.

Example 3.4. Shannon calculated [18] the per-letter unconditional entropy of the English language to be ~ 4.8 bits. However when conditioned on the previous word, the entropy reduced to ~ 2.1 bits. The optimal conditional encoding is more than twice as short as the unconditional alternative.

One solution is to implement block prefix encodings: we minimise the prefix code for $|\mathcal{X}|^N$ for a specified length N. However this grows exponentially computationally expensive with length and does not solve the problem.

Example 3.5. Consider predicting the final word in this pair of sentences: 'Alice taught Bob information theory. Bob was glad that he had learnt about information theory from _____'.

It is clear to us that this would be Alice. However to reflect the true low entropy of this, we would need a block encoding spanning across multiple sentences. Assuming we consider the English alphabet and a space character, we would need to compute

$$27^{98} \approx 2 \times 10^{140}$$

conditional frequencies to determine the precisely optimal prefix code. This is fairly impractical, since it would take orders of magnitude longer than the age of the universe to perform that calculation.

We now briefly present an alternative approach known as arithmetic coding, which avoids the issue of a minimum code length per-bit, requires less computations than a block prefix code and achieves near-optimal performance. The scheme is worth admiring in its own right, and reiterates the connection between prediction and compression. Whereas for prefix codes we saw that the code length provided an implicit likelihood, in arithmetic coding our description of a sequence *is* the sequence's likelihood.

Example 3.6 (Arithmetic Coding)**.** Suppose we have a finite alphabet $\mathcal{A} = \{A, B, C, D\}$ of symbols and wish to encode some sequence of them, which need not be an i.i.d. draw. Arithmetic coding works by identifying each sequence with an interval on $[0, 1]$.

We have some encoder that provides a predictive distribution over the symbols at each step. For the first symbol in the sequence, we partition $[0, 1]$ into subintervals, each one corresponding to a symbol and with size equal to the predicted probability of that symbol (see Figure 3.2). Suppose the first symbol was B. We then calculate an updated conditional predictive distribution $p(\_|B)$. We partition the subinterval corresponding to B as before, this time with lengths proportional to the conditional probabilities or equivalently equal to the joint likelihood of $(B, \_)$. Importantly, at each step there is a systematic ordering to the intervals which is specified beforehand.

Continuing in this manner, we can describe any sequence of events up to some fixed time by the interval it uniquely represents on $[0, 1]$, with the interval's length equal to the probability of the sequence.

We now use this to encode the sequence. We consider representing intervals in $[0, 1]$ in base 2. For example we write the string 10 to correspond to the interval $[0.10, 0.11) = [2^{-1}, 2^{-1} + 2^{-2}) = [1/2, 3/4)$, just as 0.25 corresponds to $[0.25, 0.26)$. For a given sequence, we encode it by the shortest binary string entirely contained within the interval. See again Figure 3.2 for an example. This ensures unique and efficient representations for every possible sequence of events.

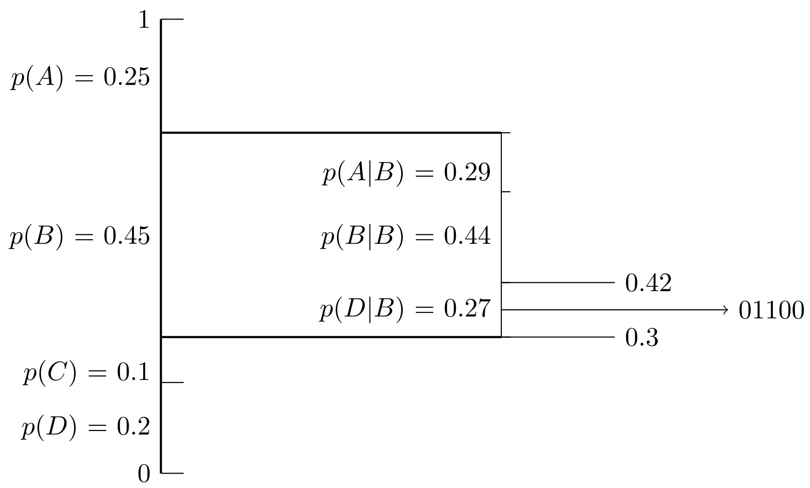

Figure 3.2: An arithmetic encoding of the sequence (B, D) gives string 01100. That $p(C|B) = 0$ is shown implicitly in second step.

To recover the sequence of events from the binary string, we use a decoder with the same predictive distribution. We calculate the probabilities of each symbol, and identify them with intervals just as the encoder: recall the intervals are assigned systematically, so their intervals on $[0, 1]$ are the same. We continue along the binary string until it is entirely contained within one of these intervals, at which point we know the first symbol. We then calculate the conditional distribution over the symbols, systematically assign the likelihoods to intervals and repeat until we reach the end of the string. The longest sequence of symbols whose corresponding interval contains the binary string entirely is then our source sequence.

Remark. Note that whereas for block prefix codes, we must consider $|\mathcal{X}|^n$ sequences to encode a given length-n sequence, for arithmetic coding we need only evaluate $n|\mathcal{X}|$ conditional likelihoods.

Lemma 3.9. *With respect to the probabilistic model of the encoder, $\mathcal{H}$, arithmetic encoding ensures a code length of at most 2 greater then than the optimal code length $h(x|\mathcal{H}) = -log_2 p(x|\mathcal{H})$.*

Proof. Suppose we attempt to maximise the length of the binary string encoding a given sequence likelihood. To do this, we consider where the interval I corresponding to its likelihood could be placed on $[0, 1]$ by the encoder. Suppose the sequence has likelihood equal to some negative integer power of 2. Then there exist intervals specified by binary strings of the same length as I. Consequently the longest string we can achieve is $h(x|\mathcal{H}) + 1$: we could place I to lie in two binary intervals of the same size as it, but then at least half of I must be in one of the binary intervals. Then at least half of one of the two binary intervals is contained in I, since the intervals are the same size as I. The binary string of length $h(x|\mathcal{H}) + 1$ specifying this interval is entirely contained in I.

Now suppose the likelihood is not some negative integer power of 2. Then at best I can be encoded by a binary interval of length $h(x|\mathcal{H}) + 1$: the interval specified by any shorter binary string is larger than I and so cannot be contained in it. However similarly to above, we can force a longer encoding by placing I to lie in two of these binary intervals. By the same argument, though, there must exist a binary string of length $h(x|\mathcal{H}) + 2$ such that the interval specified by it is entirely contained in I. □

Corollary 3.10. *Arithmetic coding achieves a code length at most 2 bits more than the optimal coding length for lossless compression under the predictive distribution.*

Proof. By Corollary 3.6, we saw how entropy remains the fundamental limit for non-i.i.d. sequences. For an arithmetic compression C of a sequence of random variables X^N, we therefore have

$$H(X^N) \leq L(C, X^N) < \sum_{x^n \in \mathcal{X}^N} p(x^n)\big(h(x^n) + 2\big) = H(X^N) + 2.$$

□

Remark. Since this bound does not depend on the length of a sequence, this implies exceptional efficiency for long sequences - especially in comparison to symbol coding methods.

Although this reduces the computations required, cases such as Example 3.5 still present an issue. In order to compress such strings effectively, we need the encoder to be a sufficiently good approximator of long conditional distributions. Learning such distributions is the entire challenge of language modelling - one of the key motivating factors behind the transformer architecture that underpins all of today's large language models was this problem of retaining context - and the success in that field has lead to the language models of today being practical versions of arithmetic encoders [4]. A good introductory lecture on the motivation and nature of the transformer architecture is given in [15].

3.2.4 Summary

Our discussion on compression has been carried out with three main goals. The first was to underscore the natural emergence of Shannon's framework, the second to justify the interpretation of entropy as a fundamental measure of the complexity of a distribution as in Proposition 1.7 and the third to show the equivalence between prediction and compression. Consider the Source Coding Theorems (Theorems 3.2 and 3.5) with respect to the first two and Theorem 3.5 and Example 3.6 with respect to the third.

As we begin to work towards an information-theoretic perspective of machine learning, the dual view of a predictive model that this equivalence allows will prove incredibly useful, both intuitively and theoretically. In order to gain an even more general perspective on the relation between information and compression, we generalise our discussion to lossy compressions. The study of the limits of such compressions is the focus of rate-distortion theory.

Chapter 4

Rate-Distortion Theory

Shannon's Source Coding Theorems provided us with precise bounds on the length of any decodable lossless compression. We now look to generalise this discussion, by considering the limits of compression when permitted a level of loss. The central task of this chapter will be to characterise the rate-distortion function, which describes the optimal encoding for a given level of acceptable imprecision. We shall see that the mutual information between a source and its encoding, rather than the entropy of the source alone, is now the fundamental limit (Theorem 4.1). Our argument shall mirror previous approaches, relying on the notions of typicality and random codings to find the asymptotic limits of lossy compressions, and is adapted from [1].

4.1 Lossy Compression

The problem of lossy compression appears in a diverse range of domains, and is a foundation to many of the interactions we have with data today. Two simplified mechanisms of processes that involve lossy compression are the following:

Example 4.1 (PCA and Image Compression)**.** Suppose we have taken a high-definition photograph and want to store it within a given file size. How can we best do this so that the compressed image looks as good to us as possible?

One possibly-familiar algorithm which can achieve a lossy compression in this instance comes from principal component analysis (PCA). Recall from linear algebra that PCA identifies, in order, the directions within which the highest variance is contained.

We break an image up into blocks of pixels, say of size $n \times n$. For an RGB photo, we then view the intensities within each block as points within a $3n^2$-dimensional space. By retaining only the top-k directions of variance and recombining, we get an image where the majority of the variance - and therefore hopefully detail - within the image remains, but described in potentially a significantly-smaller k-dimensional subspace. In particular, we can represent each block as a liner combination of the k principal components. We can then vary k as the permissible file size varies. [6]

Example 4.2 (Music and the Fourier Transform)**.** Audio, unlike images, does not have a clear discrete representation. So how can we transmit and store music?

This question touches on two problems: that of storing continuous sources (for example, real numbers) more generally, and that of storing continuous functions in particular. To achieve the latter, we take inspiration from the study of Fourier series.

As a brief introduction, for (sufficiently nice) periodic functions, we can decompose them into an infinite series with terms given by the frequencies of sines and cosines: $\{\sin(k\theta), \cos(k\theta) : k \in \mathbb{Z}\}$. For a periodic sound wave, these terms can represent the different frequencies present, and their coefficients the corresponding amplitudes.

The Fourier Transform extends this idea to non-periodic functions, and means that we can recover the frequencies and amplitudes that formed a section of music, for example, from their superposition.

To achieve a lossy compression of the source, we can discard the coefficients for any frequencies which lie outside the range of human hearing, or whose amplitude is sufficiently small [9]. Note a difference here to the previous example: whereas there, we lose some detail which may potentially 'improve' the image, here we are simply removing unnecessary information stored in the source.

As a more general example of learning and lossy compression:

Example 4.3. When taking notes during a lecture or talk, we tend not to transcribe the content word for word. This is effectively a lossy compression, too. We look to record all relevant information as efficiently as possible.

In the remainder of this chapter, we formalise the problem of lossy compression and describe the theoretical limits. We will show that the fundamental bound on the quality of a compression is not the entropy of the source variable - as it was in lossless - but the mutual information shared between the source and the output.

4.2 Defining the Rate-Distortion Problem

Given a constraint on either the average code length or the quality of the reconstruction, we look to find the limits of performance with respect to the other factor. Again, consider a discrete source X with alphabet $\mathcal{X} = \{x_1, \ldots, x_n\}$ and distribution $p(X)$. We look to compress $\mathcal{X}$ into our output alphabet $\widehat{\mathcal{X}} = \{\hat{\mathrm{x}}_1, \ldots, \hat{\mathrm{x}}_m,\}$, where generally $m << n$. In particular, we look to compress sequences of i.i.d. samples $x^n \in \mathcal{X}^n$ into output sequences $\hat{\mathrm{x}}^n$ from $\widehat{\mathcal{X}}^n$. The compression $\hat{x}^n$ of x^n is the known as the quantization of x^n.

We consider the length of the encoding in terms of its rate:

Definition 4.1 (Rate)**.** An encoding scheme has rate R if each source sequence has a representation of length nR bits.

Different functions are appropriate for evaluating a compression scheme under different contexts, so we unify all such measures through the concept of a distortion function:

Definition 4.2 (Distortion Functions)**.** A distortion function $d : \mathcal{X} \times \widehat{\mathcal{X}} \to \mathbb{R}^+$ measures the quality of a compression scheme. If

$$d_{\max} := \max_{x \in \mathcal{X},\ \hat{\mathrm{x}} \in \widehat{\mathcal{X}}} d(x, \hat{\mathrm{x}}) < \infty,$$

then it is said to be a bounded distortion function.

We extend this to sequences $x^n, \hat{\mathrm{x}}^n$ by averaging the distortion over the sequence:

$$d(x^n, \hat{\mathrm{x}}^n) = \frac{1}{n}\sum_{i=1}^{n} d(x_i, \hat{\mathrm{x}}_i).$$

Similar to distortion functions themselves, this extension is not necessarily the only choice. For example if we are concerned with achieving a consistently-faithful compression we could consider the maximum distortion over the sequence. However we stick with averaging now, aligning with the general information-theoretic approach of considering performance in expectation.

Example 4.4 (Common Distortion Functions)**.** The Hamming distortion is a binary indicator of whether the compression is exact:

$$d(x, \hat{x}) = \begin{cases} 0 & \hat{x} = x \\ 1 & \hat{x} \neq x. \end{cases}$$

Any norm on a space also defines a distortion function. Consider the need to compress real numbers to floating point numbers. A relevant distortion function might then be the absolute distance between a point and its compression:

$$d(x, \hat{x}) = |x - \hat{x}|.$$

We write the exact coding scheme of a lossy compression algorithm as the composition of an encoding function and a decoding function. We consider $(2^{nR}, n)$-rate distortion codes (rate R encoding for an alphabet with n source symbols), which have encoder $f : \mathcal{X}^n \to \{1, \dots, 2^{nR}\},\ R \in \mathbb{Z}$, and decoder $g : \{1, \dots, 2^{nR}\} \to \widehat{\mathcal{X}}$.

As we usual, we are concerned with the average performance so we take the expectation over the alphabet of $\mathcal{X}^N$:

Definition 4.3 (Distortion Associated with a Code)**.** The distortion associated with a $(2^{nR}, n)$-rate code of functions (f_n, g_n) is given by

$$D = \mathbb{E}\left[\ d\big(X^N, g_n\big(f_n(X^n)\big)\big)\right],$$

where we take the expectation over the distribution of X^n.

Example 4.5 (Zero Distortion Compressions)**.** In the examples of distortion functions given above, the distortion associated with any code can be zero if and only if the code is an exact reconstruction. However this is not always the case.

Recall our description of the Fourier Transform to compress audio. A valid compression function for audio signals might be one that is zero when the frequencies within human hearing range are perfectly recovered. For this function, only a subset of the information stored in our original input is relevant to the distortion function. That which is irrelevant can be discarded and so there can exist a 0-distortion, lower-entropy compression.

For a given source, we characterise pairs on the rate-distortion plane by whether or not a code can achieve the pairs' constraints in the limit. In particular, we are concerned with whether the distortion constraint can be aymptotically achieved given the rate constraint:

Definition 4.4 (Achievable)**.** A rate-distortion pair (R, D) is achievable if there exists a sequence of $(2^{nR}, n)$-rate distortion codes (f_n, g_n) such that

$$\lim_{n\to\infty} \mathbb{E}\left[\ d(X^N, g^n(f^n(X^n)))\right] \leq D.$$

Definition 4.5 (Rate-Distortion Region)**.** The rate-distortion region is the closure of the set of all achievable pairs (R, D).

Example 4.6. Suppose we have some source X and consider the vertical slice of the rate-distortion region given by pairs with rate $R > H(X)$. Then $(R, 0)$ is in the rate-distortion region for any source, since by the Source Coding Theorem (R, ε) is achievable for all $\varepsilon > 0$.

If X is uniformly distributed, then $(H(X), 0)$ is not only in the rate-distortion region, but is also achievable: $2^{H(X)} = |X|$ so we do not need to compress the sequence at all. If not, then the achievability of $(H(X), 0)$ depends on the distortion function.

Definition 4.6 (Rate-Distortion Function)**.** The rate-distortion function $R(D)$ is the infimum of rates R such that (R, D) is achievable for a given distortion D. Equivalently, it is the infimum of the rate distortion region at distortion D.

Definition 4.7 (Distortion-Rate Function)**.** The distortion-rate function $D(R)$ is defined similarly as the infimum of distortions D such that (R, D) is achievable for a given R.

Remark. In some sense, the distortion-rate function seems more natural to consider with Definition 4.4 in mind. We shall soon show that the distortion-rate function is in fact the inverse of the rate-distortion function within the right region. We follow the literature and work with the rate-distortion function.

4.3 The Information Rate-Distortion Function

The rate-distortion function provides a more detailed image of the distribution of a source than the entropy. In effect we go from the one-dimensional distortion-0 achievable interval (for a distortion function which requires exact reconstruction for zero distortion) to a two-dimensional rate-distortion plane.

The central theorem of this chapter is that we can express the rate distortion function in terms of the mutual information between source and quantization. We first define the information rate distortion function, a measure of the minimum relevant information that must be shared between input and output to achieve a given distortion, then prove that this is equivalent to the rate-distortion function.

Definition 4.8 (Information Rate Distortion Function)**.** For a source X with distribution $p(X)$ and distortion function $d(x, \hat{\mathrm{x}})$, let

$$S_D = \{p(\hat{\mathrm{x}}\,|x) : \sum_{x,\hat{\mathrm{x}}} p(x)p(\hat{\mathrm{x}}\,|x)d(x, \hat{\mathrm{x}})\ \ \leq D\}.$$

In other words, S_D contains the conditional distributions over $\hat{\mathrm{x}}$ such that the joint distributions over $(x, \hat{\mathrm{x}})$ satisfy the distortion constraint.

The information rate distortion function is then given by

$$R^{(I)}(D) := \inf_{S_D} \mathbb{I}(X; \hat{X}).$$

For a finite alphabet, in fact the infimum is achieved [2] and so we can write

$$R^{(I)}(D) = \min_{S_D} \mathbb{I}(X; \hat{X}).$$

Remark. Considering the discussion so far, it might seem strange to have a conditional distribution over the output alphabet when conditioned on the input, instead of a non-random function. We should want to map each point to the one with which it has the lowest distortion.

A sharp distribution, where $p(\hat{X}|X)$ is deterministic for each given realisation of X, can be written in this way. So an optimal mapping which minimises the distortion for each $x \in \mathcal{X}$ can be written in this way.

Moreover, this is in fact a useful extension as it also allows us to consider *noisy* situations, where for example the encoding is non-deterministic.

We shall working towards proving the following:

Theorem 4.1 (Information Rate-Distortion Equals Rate-Distortion)**.** *For an iid source X and bounded distortion function $d(x, \hat{\mathrm{x}})$,*

$$R(D) = R^{(I)}(D).$$

We first show that the rate-distortion function is bounded below by the information rate-distortion function, then argue that any rate greater than the information rate-distortion function is achievable.

Remark. The rate-distortion - information rate distortion equivalence also exists for well-behaved continuous sources and for unbounded distortion functions [1]. We shall rely on this fact in section 5, however here we only show the equivalence for discrete sources.

4.4 Rate $R^{(I)}(D)$ compressions have Distortion at least D

To prove Theorem 4.1, we first show that the information rate distortion function is convex.

Lemma 4.2. *$R^{(I)}(D)$ is a convex function of D.*

Proof. Consider two pairs $(R_1^{(I)}, D_1)$, $(R_2^{(I)}, D_2)$ on the information rate distortion curve. Suppose these pairs are attained by joint distributions $p_1(X, \widehat{X}), p_2(X, \widehat{X})$. Consider for $\lambda \in [0, 1]$ the mixture distribution

$$p_\lambda(X, \widehat{X}) = \lambda p_1(X, \widehat{X}) + (1 - \lambda) p_2(X, \hat{X}).$$

Since the distortion function is an expectation over the distribution, it is a linear function of the distribution and therefore

$$D_\lambda(X, \hat{X}) = \lambda D_1(X, \hat{X}) + (1 - \lambda) D_2(X, \hat{X}).$$

Now

$$\begin{aligned} R^{(I)}(D_\lambda) &\le \mathbb{I}_{p_\lambda}(X; \widehat{X}) && (a) \\ &\le \lambda\, \mathbb{I}_{p_1}(X; \widehat{X}) + (1 - \lambda)\, \mathbb{I}_{p_2}(X; \widehat{X}) && (b) \\ &= \lambda R^{(I)}(D_1) + (1 - \lambda) R^{(I)}(D_2), && (c) \end{aligned}$$

where (a) follows from the definition of the information rate distortion function as the minimum over all distributions which achieve distortion at most D_λ, (b) from the convexity of mutual information (Theorem 2.13) and (c) since the pairs lie on the information rate distortion curve. $\square$

With this result, we turn to proving the lower bound on the rate-distortion function for a given compression rate.

Theorem 4.3. *For a given distortion D, if $R < R^{(I)}(D)$, then the rate-distortion pair (R, D) is not achievable. Equivalently, any code with distortion at most D must have rate at least $R^{(I)}(D)$.*

Proof. We work to prove the second framing in the theorem statement: for any code with distortion D and rate R, $R^{(I)}(D) \leq R$. Suppose we have a $(2^{nR}, n)$-rate distortion code defined by the pair (f_n, g_n), and with distortion D over $\mathcal{X}$. Then we have the following:

$$\begin{aligned}
nR &\geq H(f_n(X^n)) && (a)\\
&\geq H(f_n(X^n)) - H(f_n(X^n)|X^n) && (b)\\
&= \mathbb{I}(X^n; f_n(X^n))\\
&\geq \mathbb{I}(X^n; \widehat{X}^n) && (c)\\
&= H(X^n) - H(X^n|\widehat{X}^n),
\end{aligned}$$

where (a) follows from the upper bound on entropy (Lemma 2.17) given that $f_n(X^n))$ has range at most 2^{nR}, (b) since discrete entropy is non-negative and (c) from the data processing inequality (Theorem 2.20), since $X^n \perp \widehat{X}^n | f(X^n)$ by the construction of the encoder.

Recalling one interpretation of mutual information, we see that we have simply upper-bounded the number of bits of information which our intermediate encoding $f_n(X^n)$ contains about X^n by nR: the number of bits of information $f_n(X^n)$ has in total.

Continuing,

$$\begin{aligned}
\mathbb{I}(X^n; \widehat{X}^n) &= \sum_{i=1}^{n} H(X_i) - H(X^n|\widehat{X}^n) && (d)\\
&= \sum_{i=1}^{n} H(X_i) - \sum_{i=1}^{n} H(X_i|\widehat{X}^n, X_{i-1}, \ldots, X_1) && (e)\\
&\geq \sum_{i=1}^{n} H(X_i) - \sum_{i=1}^{n} H(X_i|\widehat{X}_i) && (f)\\
&= \sum_{i=1}^{n} \mathbb{I}(X_i; \widehat{X}_i),
\end{aligned}$$

with (d) coming from the entropy definition of mutual information (Lemma 2.5), the independence of the sequence terms, the fact that entropy is linear for independent random variables (Corollary 2.4) and the chain rule for entropy (Lemma 2.3), (e) from the chain rule again and (f) from the fact that conditioning reduces entropy (Lemma 2.18).

Remark. As a slight digression, this string of inequalities is a fascinating result in itself. We have shown that given a constraint on total code length, the amount of information we can transmit about even i.i.d. random variables is strictly greater if we encode them jointly than if we encode each one separately (even if independent, the removal of the X_is and their corresponding quantizations removes information about the coding scheme that makes observing $\hat{X}_i$ more informative).

Now, returning to the proof, we keep chaining inequalities:

$$\begin{aligned}
&\geq \sum_{i=1}^{n} R^{(I)}\left(\mathbb{E}\left[d(X_i, \widehat{X}_i)\right]\right) && (g) \\
&\geq nR^{(I)}\left(\frac{1}{n}\sum_{i=1}^{n} \mathbb{E}\left[d(X_i, \widehat{X}_i)\right]\right) && (h) \\
&= nR^{(I)}\left(\mathbb{E}\left[d(X^n, \widehat{X}^n)\right]\right) \\
&= nR^{(I)}(D) && (i),
\end{aligned}$$

with (g) by the definition of the rate distortion function (Definition 4.6), (h) from the convexity of the information rate distortion function as a function of distortion (Lemma 4.2) and the definition of convexity (Definition 2.9) and finally (i) from our initial assumption on the distortion of our code. □

4.4.1 Aside: Relating Information Rate Distortion and Distortion-Rate

Jumping ahead of ourselves briefly, assume that the rate distortion function is equivalent to the information rate distortion function. Then we can begin to construct an image of the general behaviour of the rate distortion function. First, we note that it agrees with our previous work on lossless compression:

Example 4.7. Assume that the rate distortion function is equivalent to the information rate distortion function. Assume that our distortion function is zero if and only if the input sequence is exactly reconstructed (for example, Hamming distance). Then $R(0) = H(X)$, our limit on lossless compression, since $S_0 = p(X)$ only (Definition 4.8) and $\mathbb{I}(X : X) = H(X)$.

More generally:

Lemma 4.4. *Assume that the rate distortion and information rate distortion functions are one and the same. Then $R(D)$ is a non-increasing function in D, with bounds $0 \leq R(D) \leq H(X)$.*

Proof. As D increases, we are taking the minimum over nested sets increasing in size. Therefore it must be non-increasing. The minimum mutual information when $D = 0$ depends on the distortion function. In the most restrictive case, the distortion function could require an exact reconstruction of X, which as discussed gives $R(0) = H(X)$. The non-negativity of mutual information (Corollary 2.9) then provides the lower bound. □

With the convexity result of Lemma 4.2 in hand, we can outline the subset of the plane within which the rate distortion function has an inverse.

Lemma 4.5. *Assume that the rate distortion function is equal to the information rate distortion function. Assume that the distortion function takes on values in the range* $0 \leq D \leq d_{\max} \leq \infty$.

Let $R_0 \leq H(X)$ *be the minimum rate required such that* $(R_0, 0)$ *is in the rate-distortion region. Let* $R_{\max}$ *be the supremum of the rates such that* $(R_{\max}, d_{\max})$ *is in the rate-distortion region.*

Then the rate distortion function restricted to the range $(R_{\max}, R_0]$ *is invertible with range* $[0, \infty)$ *if the distortion function is unbounded, or range* $[0, d_{\max})$ *otherwise.*

Proof. We look to use the convexity of the information rate distortion function (Lemma 4.2) to show that the function is bijective in this region.

First, note that it is surjective because $\mathbb{I}(X; \hat{X})$ is continuous with respect to $p(\hat{X}|X)$. If the information rate distortion function is constant, then the lemma is vacuous so let us assume that it is somewhere non-constant. To show that it is injective, we look to show that it is strictly decreasing. We have already shown that it is non-increasing (Lemma 4.4), so it remains to show that it is nowhere constant.

Note that the minimum $R_{\max}$ is uniquely attained by construction. Suppose that the function was constant with rate R on some interval $[d_1, d_2]$. Then the line segment between (R, d_1) and $(R_{\max}, d_{\max})$ would line under R at d_2, contradicting the convexity of the function (Lemma 4.2). Hence the information rate distortion function is nowhere constant. It is therefore strictly decreasing so bijective within the region, meaning it admits an inverse. □

4.5 $(R^{(I)}(D), D)$ Lies in the Rate-Distortion Region

When proving Shannon's source coding theorem, we used the notion of typicality - the fact that almost all sufficiently long sequences become almost equally likely - to argue that an arbitrary encoding of just those sequences in the typical set would eventually cover almost all the probability mass over the distribution of sequences. We extend this notion of typicality to a pair of random variables.

Definition 4.9 (Jointly Typical Set)**.** Let $p(X, \widehat{X})$ be a joint distribution over $\mathcal{X} \times \widehat{\mathcal{X}}$. For $\varepsilon > 0$, a pair $(x^n, \hat{\mathrm{x}}^n)$ is jointly typical if

$$\begin{aligned} |-\frac{1}{n}\log p(x^n) - H(X)| &< \varepsilon, \\ |-\frac{1}{n}\log p(\hat{x}^n) - H(\hat{X})| &< \varepsilon, \\ |-\frac{1}{n}\log p(x^n, \hat{x}^n) - H(X, \hat{X})| &< \varepsilon. \end{aligned}$$

We denote the jointly typical set over $p(X, \widehat{X})$ by $A_\varepsilon^{(n)}$.

Definition 4.10 (Distortion Typical Set)**.** Consider the same situation as above. Suppose we also have a distortion function $d : \mathcal{X} \times \widehat{\mathcal{X}} \to \mathbb{R}^+$. For $\varepsilon > 0$, a distortion ε-typical set is a jointly typical set with the additional condition that

$$|d(x^n, \hat{x}^n) - D(X, \hat{X})| < \varepsilon.$$

The distortion typical set over $p(X, \widehat{X})$ for distortion function d is denoted $A_{d,\varepsilon}^{(n)}$. Clearly $\mathrm{A}_{\mathrm{d},\varepsilon}^{(\mathrm{n})} \subseteq A_\varepsilon^{(n)}$.

As with an ε-typical set of sequences from a source X, the probability mass concentrates within both the jointly typical and distortion typical sets as $n \to \infty$. In particular, almost all input-output pairs (again, using 'almost all' in terms of likelihood) have almost equal distortions:

Lemma 4.6. *Let n pairs $(X_i, \hat{X}_i)$ be drawn i.i.d. $\sim p(x, \hat{x})$. Then*

$$P(\mathrm{A}_{\mathrm{d},\varepsilon}^{(\mathrm{n})}) \to 1 \text{ as } n \to \infty.$$

Proof. Consider the four inequalities which define the distortion typical set. In each one, note that the left hand terms are the absolute difference between an average of n i.i.d. random variables and their expected value.

For each one of these inequalities individually, it therefore follows from the weak law of large numbers that the probability of a sequence satisfying the inequality goes to 1. By the union bound, the probability that a single inequality is not satisfied is less than or equal to the sum of the individual probabilities and therefore goes to 0. Thus the probability of all four being satisfied goes to 1. □

Notice that while the ε-typical set characterises the joint and the marginals, it does not characterise the conditional distribution. It will prove useful later to relate the behaviour of the conditional distribution to that of the marginal distribution for $\widehat{X}$. We focus on the relationship within the distortion typical set:

Lemma 4.7. *For all pairs $(x^n, \hat{x}^n)$ in the jointly typical set $\mathrm{A}_{\varepsilon}^{(\mathrm{n})}$,*

$$p(\hat{x}^n) \geq p(\hat{x}^n|x^n)2^{-n(\mathbb{I}(X;\hat{X})+3\varepsilon)}.$$

Proof. We use the three properties that describe the jointly typical set to bound the marginal and joint probabilities of all $(\hat{x}^n, x^n) \in \mathrm{A}_{\varepsilon}^{(\mathrm{n})}$:

$$\begin{aligned}
p(\hat{x}^n|x^n) &= p(\hat{x}^n)\frac{p(x^n, \hat{x}^n)}{p(x^n)p(\hat{x}^n} \\
&\leq p(\hat{x}^n)\frac{2^{-n(H(X,\hat{X}-\varepsilon)}}{2^{-n(H(X)+\varepsilon)}2^{-n(H(\hat{X})+\varepsilon)}} \\
&= p(\hat{x}^n)2^{-n(-\mathbb{I}(X;\hat{X})-3\varepsilon)},
\end{aligned}$$

where the last line comes from the result

$$\mathbb{I}(X;,\hat{X}) = H(X) + H(\hat{X}) - H(X,\hat{X}).$$

Multiplying through provides the stated form of the lemma. □

Corollary 4.8. *The same bounds holds for all $(x^n, \hat{x}^n) \in \mathrm{A}_{\mathrm{d},\varepsilon}^{(\mathrm{n})}$.*

Finally we state the following inequality, which will allow us to simplify our bounds later. The proof follows from some basic calculus, and we omit it here. It can be found in [1].

Lemma 4.9. *For $0 \leq x, y \leq 1,\ n > 0$,*

$$(1 - xy)^n \leq 1 - x + e^{-yn}.$$

With these results in hand, we can now prove the achievability of the information rate-distortion function. This, along with Theorem 4.3, will allow us to conclude that the rate-distortion function is indeed the information rate-distortion function.

Theorem 4.10 ($R^{(I)}(D) \geq R(D)$)**.** *For any source X with bounded distortion measure d, the rate-distortion pair $(R^{(I)}(D), D)$ lies in the rate-distortion region.*

Proof. Let $X^n = (X_1, \ldots, X_n)$ be i.i.d. drawn from $X \sim p(X)$. Let $d(x, \hat{x})$ be the bounded distortion measure. Let us fix some distortion value D, and let $p(\widehat{X}|X)$ be the distribution corresponding to an encoding $\hat{X}$ satisfying $R^{(I)}(D) = \mathbb{I}(X, \hat{X})$. We aim to show that in the limit, even a random coding scheme is achievable if $R > R^{(I)}(D)$.

To this end, we first generate a random set of strings $\mathcal{C}$ by drawing 2^{nR} i.i.d. sequences of $\widehat{X}^n \sim \Pi_{i=1}^n p(\widehat{X}_i)$ and indexing these codewords by $w \in \{1, \ldots, 2^{nR}\}$ to create a codebook.

Our encoding procedure for a string x^n is as follows: choose $\varepsilon > 0$. Send x^n to the index corresponding to the string $\hat{x}^n$ with which it has the lowest distortion. If more than one such strings exist, then arbitrarily send x^n to the one with the lowest index. The decoded string is then simply the corresponding string $\hat{x}^n$ indexed by the w that x^n has been mapped to.

Consider the distortion over this codebook. In particular, consider the distortion in the following two cases. When there exists an $\hat{x}^n$ such that $(x^n, \hat{x}^n) \in \mathrm{A}_{\mathrm{D},\varepsilon}^{(\mathrm{n})}$,

$$d(x^n, \hat{x}^n) < D + \varepsilon.$$

The probability of such an event occurring is at most 1.

If there does not exist such an $\hat{x}^n$, then the minimum distortion between x^n and all other strings $\hat{x}^n$ can be as great as the maximum distortion over $\mathcal{X}^n \times \widehat{\mathcal{X}}^n$. Since distortion is assumed to be bounded, we denote this $d_{\max}$. Denote the probability of this happening by P_ε. Combining, we get

$$E[d(X^n, \hat{X}^n)] \leq D + \varepsilon + P_\varepsilon d_{\max}.$$

If we can make P_ε sufficiently small, then for all $\delta > 0$ there exists $\varepsilon > 0$ such that the distortion of the random codebook is less than $D + \delta$ for sufficiently large n. By definition, the rate of $\widehat{X}$ will then be achievable for distortion D.

It is worthwhile to consider the dynamics of P_ε to understand the challenge here. First, for a moment, suppose our sequence length n is fixed. Considering P_ε, we see that it is a function of both ε through the distortion typical set chosen, and the rate R, which determines the size of our codebook. To get an expected distortion close to D, we need to take ε to 0. However this decreases the size of $\mathrm{A}_{\mathrm{D},\varepsilon}^{(\mathrm{n})}$ and thus increases P_ε. Increasing the rate gives a higher probability of at least one of the independently drawn codewords being within the distortion typical set for a given input x^n and so decreases P_ε.

Letting n now vary, it might initially seem like P_ε vanishing in the limit is inevitable regardless of the rate: we have shown that all of the probability mass of $(x^n, \hat{x}^n)$ concentrates in $\mathrm{A}_{\mathrm{D},\varepsilon}^{(\mathrm{n})}$ for all $\varepsilon > 0$ after all. For example, even if we our codebook had only one sequence, the probability that a random (input, output) pair would lie in the distortion typical set converges to 1.

But this is a mistake! If the (input, output) pair was sampled from the joint distribution $p(x^n, \hat{x}^n)$, then this would be the case. However we are sampling from the marginal distributions when we generate our input and our codebook strings. So cannot use (4.6) here.

Framing this more intuitively, recall that another description of mutual information, arising from its KL divergence formulation, is that it is the cost of encoding a sample generated from the joint distribution as if it was generated from the product of the marginal distribution. By its construction $p(\widehat{X}|X)$ is the distribution containing the least amount of information about X such that the distortion over input-output pairs is less than or equal to D. Therefore if we have less information about the input, we know that we can only achieve greater than the desired level of distortion.

Consequently, we look to 'compensate' for the fact we have encoded according to product of the marginals instead of the conditional. We do this by having an entire code book of fixed points, and then providing additional information about the input by identifying the point to which it is closest. For a sufficiently large rate - equivalently for sufficiently many points in the codebook - we expect that P_ε will go to zero. To find the lower bound on this rate, we must look to relate the behaviour of the marginal samples to the behaviour of the joint distribution. Thankfully, Lemma 4.9 already does just that.

Let us denote the set of all possible code books from $\widehat{\mathcal{X}}^n$ by C and let $T(c)$ denote the set of source sequences in $\mathcal{X}^n$ which have a distortion typical pair in a given code book $c \in C$. We can write P_ε as the sum over C of the probability a source sequence does not lie in $T(c)$:

$$P_\varepsilon = \sum_{c \in C} P(c) \sum_{x^n : x^n \notin T(c)} p(x^n).$$

Since both sums are finite, we can interchange the order of summation and write this as a sum over all possible source sequences. We sum the probabilities that a codebook is not distortion typical with a given source sequence:

$$P_\varepsilon = \sum_{x^n} p(x^n) \sum_{c\ : x^n \notin T(c)} P(c).$$

Now fix a sequence x^n. Let

$$t_C(x^n, \hat{x}^n) = \mathbb{1}_{(x^n, \hat{x}^n) \in T(C)},$$

so that we can write the probability of some code word $\widehat{X}^n$ being included in a codebook C given that it is not a part of a distortion typical pair with x^n as

$$P(t_C(x^n, \hat{X}^n)) = 1 - \sum_{\hat{\mathrm{x}}^n \in \widehat{\mathcal{X}}^n} p(\hat{x}^n) t_C(x^n, \hat{x}^n).$$

Consequently we see that the probability of a code book having none of its independently sampled codewords be part of a distortion typical pair with x^n is

$$\left(1 - \sum_{\hat{x}^n} p(\hat{x}^n) t_C(x^n, \hat{x}^n)\right)^{2^{nR}}$$

and therefore

$$P_\varepsilon = \sum_{x^n \in \mathcal{X}^n} p(x^n)\Big(1 - \sum_{\hat{x}^n} p(\hat{x}^n) t(x^n, \hat{x}^n)\Big)^{2^{nR}}.$$

We now apply Lemmas 4.7 and 4.9 in turn to show that this vanishes. First, we get that

$$\sum_{\hat{x}^n} p(\hat{x}^n) t(x^n, \hat{x}^n) \geq \sum_{\hat{x}^n} p(\hat{x}^n | x^n) 2^{-n(\mathbb{I}(X;\hat{X})+3\varepsilon)} t(x^n, \hat{x}^n),$$

bounding the marginal distribution by the conditional (equivalently P_ε by the joint), and using the typicality property of $A^{(n)}_{D,\varepsilon}$ (Lemma 4.6). Note that the indicator term allows us to apply Lemma 4.7 term-wise to the entire sum, since pairs not in the distortion typical set vanish regardless.

This gives

$$P_\varepsilon \leq \sum_{x^n} p(x^n)\Big(1 - 2^{-n(\mathbb{I}(X;\hat{X})+3\varepsilon)} \sum_{\hat{x}^n} p(\hat{x}^n | x^n) t(x^n, \hat{x}^n)\Big)^{2^{nR}}.$$

Lemma 4.9 then allows us to bound above the bracketed term again by

$$1 - \sum_{\hat{x}^n} p(\hat{x}^n | x^n) t(x^n, \hat{x}^n) + e^{-(2^{-n(\mathbb{I}(X;\hat{X})+3\varepsilon)} 2^{nR})}.$$

Summing over all possible source sequences x^n, we therefore get

$$P_\varepsilon \leq 1 - \sum_{x^n} \sum_{\hat{x}^n} p(x^n, \hat{x}^n) t(x^n, \hat{x}^n) + e^{-2^{n(R - I(X;\hat{X}) - 3\varepsilon)}}.$$

Notice that

$$1 - \sum_{x^n} \sum_{\hat{x}^n} p(x^n, \hat{x}^n) t(x^n, \hat{x}^n) = 1 - P\big(t(x^n, \hat{x}^n)\big) = 1 - P\big((x^n, \hat{x}^n) \in A^{(n)}_{D,\varepsilon}\big),$$

where the probability is now taken with respect to the joint distribution. By the typicality property of Lemma 4.6, this becomes arbitrarily small for n sufficiently large.

Moreover for all rates $R > \mathbb{I}(X;\widehat{X}) + 3\varepsilon$, we see that the exponential term in the bound of P_ε vanishes in n. Therefore if and only if

$$R > \mathbb{I}(X;\widehat{X}) = R^{(I)}(D),$$

for all $\delta > 0$ there exists a sufficiently small ε and a sufficiently large n such that both

$$1 - P\big((x^n, \hat{x}^n) \notin A^{(n)}_{D,\varepsilon}\big) \text{ and } e^{-2^{n(R - I(X;\hat{X}) - 3\varepsilon)}}$$

vanish and

$$D(X^n, \widehat{X}^n) \leq D + \delta.$$

□

Combining Theorems 4.3 and 4.10, we see that the information rate distortion function gives the infimum of all rates R such that (R, D) is achievable for any given distortion D. Therefore by definition, it is in fact the rate distortion function.

4.5.1 Summary

The journey to proving the equivalence of the rate-distortion function and the information rate distortion function takes us from a single characterisation of the complexity of a distribution to a much broader spectrum. Moreover by considering different distortion functions, we allow ourselves to specify the relevant aspects of a source and understand the limits on describing each of them independently. The concentration of information within a distribution and the fact that even random codings become almost optimal in the limit are insights which reappear here, proving to be deep and general features of information. With a sufficiently detailed understanding of the nature of information, we now look to apply the concepts we have discussed so far to the study of learning.

Chapter 5

An Information-Theoretic Approach to Learning

Before introducing the information-theoretic perspective we take, we first look to motivate it by describing the problem of generalisation and outlining the theory of probably approximately correct (PAC) learning, one of the most common theoretical approaches to learnability (Section 5.1). We justify our own approach by considering the issues within the PAC framework that it can address. We consider an optimal learning algorithm and consider its loss over time (Lemma 5.2, Lemma 5.3), showing the equivalence between error and information. We then turn to rate-distortion bounds on the performance of an optimal learner (Theorem 5.6), and apply these in the context of logistic regression (Section 5.4). The framework we explore is a recent one [10] [11]. The approach it takes is an intuitive extension of our discussion so far. The monograph of [10] guides most of our discussion from Section 5.2 onwards; Lemma 5.4 and Proposition 5.5 are novel.

5.1 A Brief Introduction to Machine Learning and generalisation

We focus on the problem of supervised learning. We are given pairs of input-output data and we look to learn a mapping between the input and output spaces which reflects the true data-generating process. In machine learning we aim to do so by minimising a loss function, which we choose to provide an indicator of how well our model matches the data.

Example 5.1. Suppose that the output data y is linearly related to the input data x except for some random noise:

$$y = w^T x + \varepsilon, \text{ where } \varepsilon \sim N(0, \sigma^2).$$

A linear regression model attempts to learn the vector of coefficients w from data points $\{(x_i, y_i)\}$. For a given set of weights $\hat{w}$, the model makes predictions

$$\hat{y} = \hat{w}^T x.$$

We look to find the optimal set of weights by minimising the mean squared error between predictions and true data points:

$$w^* = \underset{w\in\mathbb{R}^d}{\arg\min}\, \frac{1}{N}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2.$$

Example 5.2. Suppose that we wanted to learn a binary classifier of data. One framing of the relationship between inputs and outputs would be

$$y = \mathbb{1}_{w^t x + \varepsilon > a}, \text{ where } \varepsilon \sim N(0,1).$$

Such a classifier would consist of a weights vector $\hat{w}$ and decision boundary $\hat{a}$ to obtain predictions

$$\hat{y} = \mathbb{1}_{\hat{w}x > \hat{a}}.$$

We could use the Hamming loss to find the optimal such values:

$$(w^*, a^*) = \underset{w\in\mathbb{R}^d,\ a\in\mathbb{R}}{\arg\min}\, \frac{1}{N}\sum_{i=1}^{n}\mathbb{1}_{y\neq\hat{y}}.$$

In other words, we attempt to find the weights which minimise the proportion of incorrectly classified points.

More precisely, though, in machine learning we aim to learn the best approximation to the conditional distribution over the output space. The perspective of a best approximation is necessary since it is possible that our algorithm cannot learn the true data-generating distribution. Linear regression can specify the best linear model, but cannot learn polynomials; polynomial regression up to a fixed degree can give the best such polynomial, but cannot learn functions that decay in the limit. Both of these regressions learn a deterministic mapping. In contrast, neural networks are often used to learn a distribution over a discrete set of possible outcomes. They, too, still have practical limits on their complexity.

Example 5.3 (Language Models)**.** Recall that we mentioned in Section 3.2.3 that language models learn to provide a distribution over all possible outputs when given a string of preceding text as input. To be more specific, language models learn a function that maps from the space of inputs to the space of possible next additions.

Example 5.4 (Recommender Systems)**.** Alternatively, consider the recommendation system of a streaming service like Netflix. One part of this might be an algorithm that looks to predict how much different films will be enjoyed if recommended to a given user.

Taking the past actions of the user as input - for example which films were watched fully, which were stopped halfway through, and which were scrolled past - the algorithm would look to learn a conditional distribution over all titles of the likelihood of the user enjoying that film next.

Note that both of the loss functions in Examples 5.1 and 5.2 are also distortion functions (Example 4.2). As a rough analogy, we can view learning as a lossy compression: we can take the underlying conditional distribution as the source θ, which is then noisily encoded by a finite set of observations - the data. Our model is then the decoder, looking to learn the original distribution given this noisy representation, and it's complexity is the rate.

While a useful intuitive picture, it is somewhat misleading. Whereas the distortion function is theoretical and calculated over the entire distribution, loss functions are empirically calculated over a finite set of samples. Minimising the theoretical distortion-rate function - in other words, fixing model complexity to be say a linear model, and finding the best fit within that - gives you the best approximation to the conditional distribution. Minimising the loss function might not. For example, a look-up table which memorises each training input and its corresponding output trivially achieves zero loss in training, but the moment we examine it on a new data pair, it will be unable to provide an accurate prediction.

Such overfitting can only arise if the complexity of the model is sufficiently great: the model must be able to distort itself to match the data. In fact if the loss function measures only the prediction error, then this is an if-and-only-if complexity condition. Consider, for example, polynomials of a degree greater than the number of data points, or classifiers than can separate the plane into more sections than there are samples.

On the other hand, greater complexity increases the size of the class of models that can be learnt. Within a larger class, it becomes possible to learn models which better approximate the true distribution. Limiting complexity excessively can ensure that we never learn a good representation of the data-generating process. With this dynamic in mind, we can see that the challenge of machine learning is not to perform well on the data that we have, but to use that data to generalise well to the data we do not.

Example 5.5 (The Minimum Description Length Principle)**.** One approach to the problem of balancing complexity and overfitting is the Minimum Description Length Principle. [7] This states that best model trained on a sample of data is the one which minimises the total length of the description of the model and the description of the data when encoded according to this model.

The principle ties compression to generalisation by roughly the following argument: any regularity in the data can be used to compress it. Therefore the better the compressor, the better it must have an implicit representation of the structure of the distribution (think of our discussion of implicit probabilities in Section 3.2.2). Compression of the data implies knowledge of the structure of the data; compression of the model disincentivizes overfitting, in turn encouraging knowledge of the structure of the true distribution, rather than just the distribution of the samples.

This argument can be formalised using the tools of complexity theory [7] and is worth noting as an intuitive framework that has proven influential for discussing the problem of generalisation. Here, though, instead of discussing how to find the model that best generalises, we focus on describing and quantifying the difficult of generalisation for different learning problems.

Before turning to the information-theoretic approach that will take up most of this chapter, we describe the framework provided by the theory of Probably Approximately Correct (PAC) learning, one of the most widespread formal approaches to the problem. We do this to provide a point of comparison to the information-based perspective that follows. We shall note the similar form of results achieved.

5.1.1 PAC Learning

In the theory of PAC learning, we can consider the problem of supervised learning as follows [16]:

1. We assume that there exists an instance space $\mathcal{X}$ and an image space $\mathcal{Y}$ containing the possible values our input-output pairs might take on. We let $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ and assume that there exists some distribution $\mathcal{D}$ over $\mathcal{Z}$.

2. Given a sample S of pairs $\{z_i\} = \{(x_i, y_i)\}$, we aim to learn a prediction function $h : \mathcal{X} \to \mathcal{Y}$ from within a set of possible models $\mathcal{H}$, the hypothesis class.

3. We have some loss function $\ell : \mathcal{H} \times \mathcal{Z} \to \mathbb{R}^+$.

4. We want to select the model that minimises the risk:
$$h^* \in \arg\min_{h \in \mathcal{H}} L_{\mathcal{D}}(h),$$
where
$$\mathbb{E}_{z \in \mathcal{D}}[\ell(h, z)] =: L_{\mathcal{D}}(h)$$
is the risk associated with a model. (The minimiser might not be unique, hence $\in$ instead of $=$).

5. However this requires a calculation over the entire distribution $\mathcal{D}$, which we do not know. We therefore estimate the risk by the training error,
$$L_S(h) := \frac{1}{N} \sum_{i=1}^{n} l(h, z_i),$$
and select the predictor that minimises the sample error. This is known as Empirical Risk Minimisation (ERM):
$$ERM_{\mathcal{H}}(S) \in \arg\min_{h \in \mathcal{H}} L_S(h).$$

Note that while the prediction function is a conditional distribution, it is selected based on its minimum risk with respect to the joint distribution $\mathcal{D}$. In other words, we take into account the distribution of input points, too, and more heavily weight accuracy on those inputs which appear with higher frequency.

This is not any different from the regular supervised learning setup: for example consider a sample of data pairs where the inputs are from a discrete alphabet $\mathcal{X} = \{x_1, \ldots, x_n\}$. The greater the marginal probability of a point x_i being sampled, the higher the expected proportion of pairs in the sample that include x_i. Therefore the greater the marginal probability, the more the loss on x_i matters to the total sample loss.

Notice also that we specify a hypothesis class of potential models - for example quadratic polynomials or binary classifiers - before selecting a model within this class to avoid overfitting. As a consequence, in PAC learning, we focus on the theoretical limits of learning for given hypothesis classes. We define a class as learnable when with sufficient data, our best estimate from a sample gets arbitrarily close to the best possible estimate of the true distribution with an arbitrarily high likelihood:

Definition 5.1 (Agnostic PAC Learnable)**.** A hypothesis class $\mathcal{H}$ is said to be agnostic PAC learnable with respect to a space $\mathcal{Z}$ if there exists some function $m_{\mathcal{H}} : (0, 1)^2 \to \mathbb{N}$ and there exists some learning algorithm such that for all $\varepsilon, \delta \in (0, 1)^2$, and for every possible distribution $\mathcal{D}$ over $\mathcal{Z}$, for all i.i.d. samples from $\mathcal{D}$ of size $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$, the algorithm returns a predictor $h \in \mathcal{H}$ satisfying

$$P\big(L_{\mathcal{D}}(h) - L_{\mathcal{D}}(h^*) \leq \varepsilon\big) \geq 1 - \delta,$$

where $h^* \in \arg\min_{h\in\mathcal{H}} L_{\mathcal{D}}(h)$, and we take the probability over the distribution of data samples drawn from $\mathcal{D}$.

This is somewhat dense. To restate: for a class of models $\mathcal{H}$ to be deemed PAC learnable, we need to consider every distribution ($\mathcal{D}$). For any distribution, and for any pair of probability (δ) and distance (ε), it must be the case that for a sufficiently large sample size ($m_{\mathcal{H}}(\varepsilon, \delta)$), there exists an algorithm that will learn a model h from $\mathcal{H}$ which generalises almost as well (ε) as the best possible member h^* of $\mathcal{H}$ does, for a proportion of possible samples which have probability mass at least $1 - \delta$.

To see the need for 'with high probability' when discussing learning problems, we provide an example from a situation even simpler than learning a conditional distribution:

Example 5.6 (The Need for 'With High Probability')**.** Let us to return to the coin flip situation described back in Example 3.3. We have a coin being flipped with possible outcomes {Heads, Tails, On its side} and corresponding probabilities $\{\frac{50}{101}, \frac{50}{101}, \frac{1}{101}\}$.

Suppose that we are attempting to learn the likelihood of each outcome. We can view this as attempting to learn the distribution at a single point. Let our hypothesis class $\mathcal{H}$ be the class of probability mass functions over 3 outcomes. Note that the true distribution lies within our hypothesis class.

Suppose that we observe N independent flips of the coin, and the outcome each time is the event {On its side}. Any algorithm learning from this will not return anything close to the true distribution.

Since the probability of such a sequence being our training data vanishes exponentially in N - the likelihood is 101^{-N} - the hypothesis class can still be PAC learnable despite every algorithm failing in this situation.

The size of the necessary sample $m_{\mathcal{H}}(\varepsilon, \delta)$ depends on not only δ and ε, but also the complexity of $\mathcal{H}$. There exists a bias-complexity trade-off for $m_{\mathcal{H}}(\varepsilon, \delta)$. For a given ε and δ, it takes fewer samples for a simpler hypothesis class to, with high likelihood, find a model close to the best possible within its class, with respect to generalisation error. However as discussed earlier, while learning might be slower, the generalisation error within a more complex hypothesis class can eventually become lower.

Note the parallels between this bias-complexity trade-off and the Minimum Description Length Principle. There, instead of the size of a hypothesis class, the length of the description acts as the measure of a specific model's complexity. The description length of a data is then the measure of bias. The longer the description needed, the further the learnt distribution is from the true one. (Recall how KL divergence is both the cost of coding (Proposition 3.7) and the distance between distributions (Lemma 2.10).

Agnostic PAC learnable hypothesis classes allow us to have confidence that if we provide a large enough sample, a suitably-good training predictor will also be a suitably-good general predictor. The agnostic prefix reflects that we make no assumptions about the nature of the distribution $\mathcal{D}$ in relation to $\mathcal{H}$. In line with our earlier discussion on learning a best approximation, we do not assume that $\mathcal{D} \in \mathcal{H}$, for example.

However requiring that it must be possible to learn any distribution $\mathcal{D}$ for a hypothesis class to be learnable is a strong demand and comes at a cost. Even if our class of models cannot represent the true distribution, we would expect that it is in some way related at least: a simplified regression or classification model, perhaps. However for the class to be agnostic PAC learnable, we are forced to consider the worst-case scenarios: it must be possible to learn something close to the optimal solution no matter how pathological the distribution. This can lead to vacuous bounds, particularly for deep neural networks. Theoretically, these should require significantly more data than they are currently trained on for us to be confident that they will generalise. However in reality they show strong generalisation capabilities [25].

Whereas PAC learning considers how well classes of models can generalise from an arbitrary source, our information-theoretic approach focuses on how well any given learning algorithm can generalise from a given data-generating process. We first reframe the problem of supervised learning and define a measure of generalisation using the tools of information theory. Again, in line with the general information-theoretic perspective, we focus on performance in expectation, rather than in the worst case. We provide an exact formulation for learning and show that this is tightly bounded by our own bias-complexity equivalent. The form of this equivalent has an exact correspondence to a specific rate-distortion function.

5.2 Characterising Performance & Error

We consider the performance of learning algorithms as they are provided with new data. We have a sequence of random variables

$$(X_t, Y_{t+1})_{t\in\mathbb{Z}}$$

of input-output pairs. Suppose that Y is discrete. Let H_t denote the history up to time t,

$$(X_0, Y_1, \ldots, Y_t, X_t),$$

equivalent to the total data we have as a learner before predicting Y_{t+1}.

We assume that the sequence $(X_0, X_1, \ldots)$ is i.i.d. with respect to some known distribution. We assume that the sequence $((X_0, Y_1), (X_1, Y_2), \ldots)$ is also i.i.d. with distribution parameterised by some latent random variable θ:

$$\theta(Y_{t+1}|H_t) := P(Y_{t+1}|\theta, H_t) = P(Y_{t+1}|\theta, X_t).$$

Since we approach this from a Bayesian perspective, we model θ as a random variable with prior $p(\theta)$. We assume that the sequence of inputs $(X_0, X_1, \ldots)$ is independent of θ. Since the data-generating sequence is i.i.d., we also have that $X_t \perp \theta|(H_{t-1}, Y_t)$ as a consequence:

$$\begin{aligned}
\mathbb{I}(X_t;\theta|H_{t-1},Y_t) &= H(X_t|H_{t-1},Y_t) - H(X_t|\theta,H_{t-1},Y_t) && (a)\\
&= H(X_t) - H(X_t|\theta) && (b)\\
&= 0, && (c)
\end{aligned}$$

where (a) follows from the chain rule for mutual information (Lemma 2.16), (b) from the assumption that the sequence $((X_0, Y_1), (X_1, Y_2), \ldots)$ is i.i.d. and the fact that conditioning on an independent random variable does not reduce entropy (Lemma 2.18) and (c) from the assumption that $(X_0, X_1, \ldots) \perp \theta$ and Lemma 2.18 again.

Remark. In [10], the assumption made is not that $((X_0, Y_1), (X_1, Y_2), \ldots)$ is an i.i.d. sequence, but that it is an infinitely exchangeable one: one for which the distribution of $\big((X_t, Y_{t+1})_{t\in\mathbb{Z}}\big)$ is the same no matter the order in which we receive the data pairs. Since i.i.d. sequences of data are infinitely exchangeable - infinite exchangeability is a slightly weaker property than independence - we are choosing to limit our perspective slightly here for the sake of greater clarity.

We attempt to learn the conditional distribution of Y_t, $\theta(Y_{t+1}|H_t)$. Following the same approach that we took with lossy and lossless compression, we look to find the theoretical limits of how well any algorithm π can approximate the true distribution from a given sequence of data.

Example 5.7 (Neural Networks and Gradient Descent)**.** To provide a concrete example of a learning algorithm, let us briefly describe the backpropagation algorithm used to train neural networks. Assume that we are training a neural network as our predictor. In particular, as in Example 5.3, we are training our network to learn a function that maps from the input space to the space of distributions over the output space.

In neural networks, we can learn this function through the gradient descent optimization procedure. At each time step, we record how well our model predicts on the sample data. We then call on the backpropagation algorithm to adjust our network parameters. The algorithm calculates the gradient of the loss function with respect to each parameter and then adjusts each in the direction of lower loss (hence gradient descent).

To visualise this, consider a graph of the network parameters against loss. As we adjust our parameters, we are walking along this loss landscape. The backpropagation algorithm ensures that we are always moving downwards - in fact it sends us along the steepest downward path - so that we end up at a loss minimum.

We denote the learnt distribution at time t by $P_t = \pi(H_t)$. To investigate the theoretical limit, we focus on the average generalisation error of our algorithm. This is expressed as the average expected log-loss of our learner over the entire training trajectory, given for any $T \in \mathbb{Z}$ by

$$L_{T,\pi} = \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}_\pi[-\log P_t(Y_{t+1})].$$

We take the expectation over the space on which θ and H_t are defined. The subscript denotes that P_t is determined by $\pi(H_t)$.

Example 5.8. Returning to Example 5.4, the theoretical generalisation loss of our algorithm would be the expected predictor error over every possible history of films and actions, over every possible set of true preferences.

Note how this measures the general capabilities of a predictor, by considering all possible permutations of data and preferences, to form a general evaluation of the algorithm as a learner.

Remark. Writing the loss as an average over the course of the data sequence is slightly discordant with the usual definitions of a loss function. In particular, normally we are focused on final performance only. Taking the average encodes slightly different information about the behaviour of the algorithm. We shall see how this allows us to better relate error to learning, and provides a more detailed description of the rate of learning. First, though, we note that at all times t, there is a non-zero lower bound to the generalisation error.

Lemma 5.1. *For any learning algorithm π,*

$$L_{T,\pi} > \frac{1}{T}\sum_{t=0}^{T-1} H(Y_{t+1}|\theta, H_t).$$

Proof. Consider the algorithm π given by the true predictive distribution at each time step: $P_t = P(Y_{t+1}|\theta, H_t)$. Then

$$\begin{aligned} L_{T,\pi} &= \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[-\log P(Y_{t+1}|\theta, H_t)] \\ &= \frac{1}{T}\sum_{t=0}^{T-1} H(Y_{t+1}|\theta, H_t). \end{aligned}$$

By Gibbs' inequality (Theorem 2.8), this is the lower bound for all distributions ($E_p \log p \leq E_p \log q$ for all distributions p, q). ☐

Gibbs' inequality in fact gives us the stronger fact that the true predictive distribution uniquely achieves loss this low. So no 'true' learner - no learner that does not know θ to begin with - can achieve this minimum. For example, for a true learning algorithm, the predictive distribution for Y_1 at time $T = 0$ certainly cannot be optimal unless our prior is the true distribution. Since we average over all time steps, we can at best hope to approach the minimum asymptotically as a consequence.

A bound on the performance of any learning algorithm is given by the Bayesian posterior.

Lemma 5.2. *The optimal learning algorithm π over any horizon T is that which gives the Bayesian posterior $\hat{P}_t(Y_{t+1}) = P(Y_{t+1}|H_t)$ as the predictive distribution at each time step t. More strongly, for all $t \in \mathbb{Z}^+$, the Bayesian posterior is optimal:*

$$\mathbb{E}[-\ln \hat{P}_t(Y_{t+1})|H_t] = \min_{\pi} \mathbb{E}_{\pi}[-\ln P_t(Y_{t+1})|H_t].$$

Proof. We have

$$\begin{aligned} \mathbb{E}[-\ln P_t(Y_{t+1})|H_t] &= \mathbb{E}\left[-\ln \hat{P}_t(Y_{t+1}) + \ln \frac{\hat{P}_t(Y_{t+1})}{P_t(Y_{t+1})} \middle| H_t \right] \\ &= H(\hat{P}_t) + D_{KL}(\hat{P}_t || P_t). \end{aligned}$$

Applying Gibbs' inequality (Theorem 2.8) again, we see that the expected loss is uniquely minimised by the Bayesian posterior. ☐

Note that this result describes the optimal algorithm for any data-generating process.

Since our aim is to characterise the limit of possible performance, we focus in the remainder of this chapter on the nature of the optimal achievable estimation error. We adjust our loss function to the following:

$$\mathcal{L}_T = \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[-\ln \hat{P}(Y_{t+1})] - \mathbb{E}[-\ln P(Y_{t+1})|\theta, H_t],$$

where we subtract here the aleatoric uncertainty of the underlying distribution.

Example 5.9 (The Bayesian Posterior vs. Backpropagation)**.** One way to view Lemma 5.2 is as proof that updating the Bayesian posterior at each step is the right *level* of adjustment to our predictive distribution. However while this might theoretically be the case, calculating the Bayesian posterior at each step is often practically intractable. Unless the form of the distribution is particularly convenient, we require numerical techniques to approximate it. The process can be computationally expensive, especially for high-dimensional data.

The appeal of gradient descent (Example 5.7), in contrast, is that the adjustments can be expressed analytically and computed efficiently. It has been shown that gradient descent acts as an approximate Bayesian posterior estimator, too [13].

5.3 Connecting Learning & Information

We first show that the optimal achievable estimation error is exactly the information about θ available:

Lemma 5.3 (Optimal achievable error equals total information)**.** *For all* $T \in \mathbb{Z}^+$,

$$\mathcal{L}_T = \frac{\mathbb{I}(H_T;\theta)}{T}.$$

Proof.

$$\begin{aligned}
\mathcal{L}_T &= \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[-\ln \hat{P}(Y_{t+1})] - \mathbb{E}[-\ln P(Y_{t+1})|\theta, H_t] \\
&= \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\left[\ln \frac{P(Y_{t+1}|\theta, H_t)}{P(Y_{t+1}|H_t)}\right] \\
&= \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\left[\mathbb{E}\left[\ln \frac{P(Y_{t+1}|\theta, H_t)}{P(Y_{t+1}|H_t)}\middle|\theta, H_t\right]\right] && (a) \\
&= \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\left[D_{KL}\big(P(Y_{t+1}|\theta, H_t)||\ P(Y_{t+1}|H_t)\big)\right] && (b) \\
&= \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{I}(Y_{t+1};\theta|H_t) && (c) \\
&= \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{I}(Y_{t+1};\theta|H_t) + \frac{1}{T}\sum_{t=0}^{T-1} \mathbb{I}(X_t;\theta|H_{t-1}, Y_t) && (d) \\
&= \frac{\mathbb{I}(H_t;\theta)}{T}, && (e)
\end{aligned}$$

where (a) follows from the tower property of conditional expectation (Lemma A.8), (b) from the definition of divergence for a distribution conditioned on an event (Definition 2.12), (c) by Lemma 2.6, (d) since $X_t \perp \theta|(H_{t-1}, Y_t)$ by assumption and (e) by the chain rule for mutual information (Lemma 2.16). $\square$

We see that the optimal achievable error - the difference in loss between that of an 'omniscient' learner and that of an optimal 'true' learner - up to a given time T is exactly the amount of information about the latent variable θ that can be extracted from the data H_T.

Equating loss with information gain may appear to align two opposing metrics. However note that loss must be incurred in order to learn: we gain information about θ by adjusting from an incorrect prediction which depends on θ. Thus the amount learnt about θ is bounded by the loss with respect to predictions dependent on θ. This gives another interpretation for an optimal learner:

Lemma 5.4. *The Bayesian Posterior reduces loss at each time step by exactly the new information provided about θ.*

Proof. Let e_{T+1} denote the reduction in generalisation error between times T and $T+1$. We have

$$e_{T+1} = (T+1)\mathcal{L}_{T+1} - T\mathcal{L}_T = \mathbb{I}(H_{T+1};\theta) - \mathbb{I}(H_T;\theta).$$

$\square$

Indeed if we had defined this intuitive property as the condition for being an optimal learner, the loss at time T as in Lemma 5.3 would have immediately followed, since we get the telescoped sum

$$\frac{1}{T}\bigg(\mathbb{I}(H_0;\theta) + \big(\mathbb{I}(H_1;\theta) - \mathbb{I}(H_0;\theta)\big) + \cdots + \big(\mathbb{I}(H_T;\theta) - \mathbb{I}(H_{T-1};\theta)\big)\bigg) = \frac{\mathbb{I}(H_T:\theta)}{T}.$$

This seems to nicely square with our expectation that loss should vanish:

$$\mathbb{I}(H_T;\theta) \leq H(\theta) \text{ (Lemma (2.5)) and so } \lim_{T\to\infty} \mathcal{L}_T \leq \lim_{T\to\infty} \frac{H(\theta)}{T} = 0.$$

However this only holds in the case where the parameter space for θ is discrete. We cannot bound mutual information by differential entropy because $H(\theta|H_t)$ is not bounded below by zero. A continuous space θ is more realistically needed to represent all possible distributions - even with distributions over a discrete set, we want to be able to vary the probabilities at each point continuously over $[0, 1]$, for example - and so we must turn to different methods to characterise the behaviour of the generalisation error over time. We draw on the framework of rate-distortion theory.

5.3.1 Lossy Compression Bounds on Generalisation

For a given data sequence, we can denote the distribution learnt at time T by an algorithm by $\tilde{\theta}$, the approximation to the true parameter θ. We can evaluate the algorithm's success as a learner by comparing the time-T approximation $\tilde{\theta}$ to θ.

To define the space of possible approximations $\tilde{\theta}$, we have the same restriction on $\tilde{\theta}$ with regards to the data generating process as we do on θ, namely that $X_t \perp \tilde{\theta}|(H_{t-1}, Y_t)$ and $(X_0, X_1, \ldots, X_t) \perp \tilde{\theta}$. We also have the additional condition that $\tilde{\theta} \perp Y_t|(H_t, \theta)$ at all times t. This last condition specifies the space of random variables which do not contain any more knowledge about Y_t than can be discerned from the true distribution. Together, these conditions therefore describe all approximations $\tilde{\theta}$ that could be returned by our learning algorithm.

Next, we define the distortion function. As was mentioned briefly in Chapter 4, the information rate distortion function is equivalent to the rate distortion function for continuous sources (Remark 4.8).

For a given $\varepsilon > 0$, $\theta \in \Theta$ and data history H_t, let

$$d_t(\theta, \tilde{\theta}) = D_{KL}\Big(P(Y_{t+1}|\theta, H_t)||P(Y_{t+1}|\tilde{\theta}, H_t)\Big).$$

Then

$$\begin{aligned}
\mathbb{E}[d_t(\theta, \tilde{\theta})] &= \mathbb{E}\left[D_{KL}\Big(P(Y_{t+1}|\theta, H_t)||P(Y_{t+1}|\tilde{\theta}, H_t)\Big)\right] \\
&= \mathbb{E}\left[D_{KL}\Big(P(Y_{t+1}|\tilde{\theta}, \theta, H_t)||P(Y_{t+1}|\tilde{\theta}, H_t)\Big)\right] && (a) \\
&= \mathbb{I}(Y_{t+1}; \theta|\tilde{\theta}, H_t), && (b)
\end{aligned}$$

(a) following from our assumption on $\tilde{\theta}$ that $Y_{t+1} \perp \tilde{\theta}|(H_t, \theta)$ and (b) from Lemma 2.6.

We define our time T distortion function as the average expected divergence over all times up to T. Since $X_{t+1} \perp \theta|H_t, Y_t$, note that $\mathbb{I}(X_{t+1}; \theta|\tilde{\theta}, H_t) = 0$ for all times t. Therefore by the chain rule for mutual information (Lemma 2.16), our time T distortion function can be written as

$$\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[d_t(\theta, \tilde{\theta})] = \frac{\mathbb{I}(H_T; \theta|\tilde{\theta})}{T},$$

with the rate-distortion function at threshold ε then given by

$$H_{\varepsilon,T}(\theta) := \inf_{\tilde{\theta}\in\tilde{\Theta}_{\varepsilon,T}} \mathbb{I}(\theta; \tilde{\theta}),$$

where

$$\tilde{\Theta}_{\varepsilon,T} = \{\tilde{\theta} : \tilde{\theta} \perp H_T|\theta, \frac{\mathbb{I}(H_T; \theta|\tilde{\theta})}{T} < \varepsilon\}.$$

The condition $\tilde{\theta} \perp H_t|\theta$ is satisfied if and only if the constraints on $\tilde{\theta}$ with respect to X_t and Y_t as detailed above are satisfied.

To understand how this represents algorithm performance, consider first the distortion function. We can view this as the per-timestep information that the data provides about the true parameter that is not provided by our approximation $\tilde{\theta}$. Then as ε decreases, for example, we constrain ourselves to approximations which have absorbed more and more of the information H_T has provided about θ: we constrain ourselves to the set of more and more efficient learners.

The rate distortion function is therefore the curve describing the minimal amount of information an algorithm must have learnt about θ given that it has absorbed some level of the information from the data sequence. The Bayesian posterior is the sole member of the distortion-zero set.

Proposition 5.5 (Uniqueness of the Bayesian Posterior in the Distortion-Zero Set). *For all finite times T, the distribution $\tilde{\theta}$ learnt by the Bayesian posterior algorithm is the unique member of $\tilde{\Theta}_{0,T}$.*

Proof. Let us outline the intuition here first. The distortion-zero condition requires that $\mathbb{I}(H_T;\theta|\ \tilde{\theta}) = 0$. In other words, no information about θ can be obtained from H_T that is not contained in $\tilde{\theta}$.

Recall that we have so far shown that the Bayesian posterior is the unique optimal learner (Lemma 5.2), achieves total loss $\mathbb{I}(H_T;\theta)$ (Lemma 5.3) and acquires all the information provided from incurring a loss (Lemma 5.4) .

Consequently, we should expect that $\mathbb{I}(H_T;\theta)$ is exactly the mutual information between the Bayesian approximation to the latent variable, $\tilde{\theta}_B$, and θ. If this is the case, then by definition all of the information about θ from H_t must be contained in $\tilde{\theta}_B$. And therefore $\tilde{\theta}_B$ must have distortion zero. Since no other algorithm retains information from the data sequence as well as the posterior, it must also be the unique member. To formalise, we first show that the Bayesian posterior achieves zero distortion.

Recall that $\theta(H_t) = P(Y_{t+1}|\ \theta, H_t)$ and that the approximation $\tilde{\theta}_B$ of the Bayesian learner is the posterior distribution $P(Y_{t+1}|H_t)$. Recall also that

$$\mathbb{I}(H_T;\theta|\tilde{\theta}_B) = \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{I}(Y_{t+1};\theta|\tilde{\theta}_B, H_t),$$

as was discussed in the derivation of the distortion function, and that mutual information is symmetric (Lemma 2.5).

We look to show that

$$\mathbb{I}(H_t;\theta|\tilde{\theta}_B) = 0.$$

We use the chain rule (Lemma 2.16) to write this as

$$\mathbb{I}\left(\theta;(Y_{t+1},\tilde{\theta}_B)|H_t\right) - \mathbb{I}(\theta;\tilde{\theta}_B|H_t),$$

and see that

$$\begin{aligned}
\frac{1}{T}\sum_{t=0}^{T-1}\mathbb{I}(\theta;\tilde{\theta}_B|H_t) &= \frac{1}{T}\sum_{t=0}^{T-1} H(\tilde{\theta}_B|H_t) - H(\tilde{\theta}_B|H_t,\theta) \\
&= \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[-\log P(Y_{t+1}|H_t)] - \mathbb{E}[-\log P(Y_{t+1}|\tilde{\theta}_B,\theta,H_t)] \qquad (a) \\
&= \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{E}[-\log P(Y_{t+1}|H_t)] - \mathbb{E}[-\log P(Y_{t+1}|\theta,H_t)] \qquad (b) \\
&= \mathcal{L}_T = \frac{\mathbb{I}(H_T;\theta)}{T},
\end{aligned}$$

where (a) follows from the definition of conditional mutual information (Definition 2.13) and (b) from the assumption that $\tilde{\theta} \perp Y_{t+1}|\theta, H_t$.

In order to show that $\tilde{\theta}_B$ is distortion-zero, it remains to show that

$$\mathbb{I}(H_T;\theta) = \frac{1}{T}\sum_{t=0}^{T-1}\mathbb{I}(Y_{t+1};\theta|H_t) = \frac{1}{T}\sum_{t=0}^{T-1} I\big((Y_{t+1},\tilde{\theta}_B);\theta|H_t\big),$$

since then

$$\mathbb{I}\left(\theta; (Y_{t+1}, \tilde{\theta}_B)|H_t\right) = \mathbb{I}(\theta; \tilde{\theta}_B|H_t).$$

This follows from the fact that $\tilde{\theta}_B$ is a function of H_t and in particular that $\tilde{\theta}_B \perp \theta|H_t$. First by the chain rule for mutual information (Lemma 2.16), then by the fact that conditionally independent random variables have mutual information zero (Corollary 2.15), we have

$$\begin{aligned}\mathbb{I}\left((Y_{t+1}, \tilde{\theta}_B); \theta|H_t\right) &= \mathbb{I}(Y_{t+1}; \theta|H_t) + \mathbb{I}(\tilde{\theta}_B; \theta|Y_{t+1}, H_t) \\ &= \mathbb{I}(Y_{t+1}; \theta|H_t),\end{aligned}$$

This proves that $\tilde{\theta}_B$ is distortion-zero. Now note that everything follows as above for any other learning algorithm, with the exception that

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{I}(\theta; \tilde{\theta}_B)|H_t) = \mathcal{L}_T < \frac{\mathbb{I}(H_T; \theta)}{T},$$

by the uniqueness of the Bayesian posterior as an optimal learner (Lemma 5.2). Consequently $\mathbb{I}(\theta; \tilde{\theta}_B) < \mathbb{I}(\theta; (Y_{t+1}, \tilde{\theta}_B)|H_t)$, meaning $\mathbb{I}(H_T; \theta|\tilde{\theta}_B) > 0$. For all other algorithms, the distortion is greater than zero. ☐

In addition to this exact identity, we can bound the optimal achievable loss over the entire rate-distortion function.

Theorem 5.6 (Rate-Distortion Bounds for Learning)**.** *For all $T \in \mathbb{Z}^+$,*

$$\sup_{\varepsilon \geq 0} \min\left\{\frac{H_{\varepsilon,T}(\theta)}{T}, \varepsilon\right\} \leq \mathcal{L}_T \leq \inf_{\varepsilon \geq 0} \frac{H_{\varepsilon,T}(\theta)}{T} + \varepsilon$$

Proof. First, we look to show the upper bound. Fix $\varepsilon > 0$. For any $\tilde{\theta} \in \Theta_{\varepsilon,T}$, we have:

$$\begin{aligned}\mathcal{L}_T &= \frac{\mathbb{I}(H_T; \theta)}{T} \\ &= \frac{\mathbb{I}\left(H_T; (\theta, \tilde{\theta})\right)}{T} && (a) \\ &= \frac{\mathbb{I}(H_T; \tilde{\theta})}{T} + \frac{\mathbb{I}(H_T; \theta|\tilde{\theta})}{T} && (b) \\ &\leq \frac{\mathbb{I}(H_T; \tilde{\theta})}{T} + \varepsilon && (c) \\ &\leq \frac{\mathbb{I}(\theta; \tilde{\theta})}{T} + \varepsilon && (d),\end{aligned}$$

where for (a) we use the conditional independence of $\tilde{\theta}$ to H_T given θ and the chain rule:

$$\mathbb{I}\left(H_T; (\theta, \tilde{\theta})\right) = \mathbb{I}(H_T; \theta) - \mathbb{I}(H_T; \tilde{\theta}|\theta) = \mathbb{I}(H_T; \theta).$$

Then (b) follows from the chain rule now applied to condition on $\tilde{\theta}$, (c) from the definition of $H_{\varepsilon,T}$ and (d) from the data-processing inequality (Theorem 2.20), since $\tilde{\theta} \perp H_T|\theta$.

Since this holds for all $\tilde{\theta}$ in $\Theta_{\varepsilon,T}$ we have

$$\mathcal{L}_T \leq H_{\varepsilon,T}(\theta) + \varepsilon$$

by taking the infimum over the set. Since it also holds for all $\varepsilon \geq 0$, we can take the infimum once again to get the desired result

$$\mathcal{L}_T \leq \inf_{\varepsilon \geq 0} H_{\varepsilon,T}(\theta) + \varepsilon.$$

Turning to the lower bound, again fix $\varepsilon > 0$ and suppose that $I(H_T;\theta) < H_{\varepsilon,T}(\theta)$. Consider some $\tilde{\theta} \notin \Theta_{\varepsilon,T}$. We have

$$\begin{aligned}
\mathbb{I}(H_T;\theta) &= \sum_{t=0}^{T-1} \mathbb{I}(Y_{t+1};\theta|H_t) \\
&\geq \sum_{t=0}^{T-1} \mathbb{I}(Y_{t+1};\theta|\tilde{\theta}, H_t) && (a) \\
&\geq \varepsilon T && (b),
\end{aligned}$$

where (a) holds by the conditional independence of $\tilde{\theta}$ and Y_{t+1} and (b) follows because $\tilde{\theta} \notin \Theta_{\varepsilon,T}$.

Consequently, for all $\varepsilon > 0$, we have $\mathcal{L}_T \geq \min\left\{H_{\varepsilon,T}, \varepsilon\right\}$ and therefore, taking the supremum, we get

$$\sup_{\varepsilon \geq 0} \min\left\{H_{\varepsilon,T}, \varepsilon\right\} \leq \mathcal{L}_T.$$

□

These bounds might initially feel less exact, given the precise formulation of error (Lemma 5.3) and the identification of the Bayesian posterior within the distortion-zero set (Lemma 5.5). However they allows us to translate from error as a measure dependent on the parameter θ as in Lemma 5.3, to bounds that are a function of the data-generating process as a whole. Whereas $\mathbb{I}(H_t;\theta)$ reflects the complexity of different members of the data-generating class, we shall see that $\mathbb{I}(H_t;\theta|\tilde{\theta})$ is a relative measure of learnability and a function of the class itself.

To complete our discussion of this information-theoretic framework, we use these rate-distortion bounds to prove that optimal error goes to zero in a logistic regression process, justifying our motivation for the bounds' construction. This in fact will just be an immediate consequence of the more general image of the rate of learning that Theorem 5.6 provides.

5.4 Logistic Regression and Optimal Learning

We frame the data-generating process as follows: $X_t \sim N(0, I_d)$ is the input random vector. We observe corresponding outputs $Y_{t+1} \in \{0,1\}$ and our parameter θ is a weights vector. The weighted sum of features $\theta^T X_t$ determines the likelihood of the two outcomes:

$$Y_{t+1} = \begin{cases} 1 & w.p. \ \frac{1}{1+e^{-\theta^T X_t}} \\ 0 & o/w. \end{cases}$$

We assume a normal prior on $\theta:\ p(\theta) = N(0, I_d/d)$. Rate-distortion bounds prove easier to work with when can can focus on the error at one time step, rather than the average of them all. To this end:

Lemma 5.7 (Monotonicity of Error)**.** *Suppose $((X_t, Y_{t+1}))_{t\in\mathbb{Z}^+}$ is an i.i.d. sequence of pairs of random variables when conditioned on the underlying parameter θ. Then for all t,*

$$\mathbb{I}(Y_{t+2}; \theta|H_{t+1}) \le \mathbb{I}(Y_{t+1}|\theta|H_t).$$

Proof.

$$\begin{aligned}
\mathbb{I}(Y_{t+2}; \theta|H_{t+1}) &= h(Y_{t+2}|H_{t+1}) - h(Y_{t+2}|\theta, H_{t+1}) && (a)\\
&= h(Y_{t+2}|H_{t+1}) - h(Y_{t+2}|\theta, H_{t-1}, Y_t, X_{t+1}) && (b)\\
&\le h(Y_{t+2}|H_{t-1}, Y_t, X_{t+1}) - h(Y_{t+2}|\theta, H_{t-1}, Y_t, X_{t+1}) && (c)\\
&= \mathbb{I}(Y_{t+2}; \theta|H_{t-1}, Y_t, X_{t+1}) && (d)\\
&= \mathbb{I}(Y_{t+1}; \theta|H_{t-1}, Y_t, X_t) && (e)\\
&= \mathbb{I}(Y_{t+1}; \theta|H_t),
\end{aligned}$$

where (a) and (d) follow from Lemma 2.5, (b) from the conditional independent of Y_{t+2} to (H_t, Y_{t+1}) given θ and X_{t+1}, meaning removing any members of this sequence does not affect the entropy (Lemma 2.18), (c) by Lemma 2.18 - in particular since not conditioned on θ this is not just an equality as above - and (e) because we know that the pairs (X_t, Y_{t+1}) and (X_{t+1}, Y_{t+2}) have identical distributions given the past data sequence (H_{t-1}, Y_t) since we know that all pairs are i.i.d. under some θ. □

Lemma 5.7 allows to express error in terms of a single time step:

Corollary 5.8 (Bounding Average Error by First-Step Error)**.** *Under the same conditions as above,*

$$\frac{\mathbb{I}(H_t; \theta|\tilde{\theta})}{T} \le \mathbb{I}(Y_1; \theta|\tilde{\theta}, X_0).$$

Proof. Recall that since $X_t \perp \theta|(H_{t-1}, Y_t)$,

$$\frac{\mathbb{I}(H_t; \theta|\tilde{\theta})}{T} = \frac{1}{T}\sum_{i=1}^{T-1} \mathbb{I}(Y_{t+1}; \theta|\tilde{\theta}, H_t).$$

The proof of Lemma 5.7 still holds when conditioning on $\tilde{\theta}$, so by it the result then follows. □

We shall now state without proof a number of supplementary results which we shall need.

Proposition 5.9 ([5])**.** *We briefly mentioned in Section 2.1.1 that differential entropy is maximised by a normal random variable.*

A more precise statement of this is the following: for a random vector X with a given covariance matrix Σ, its differential entropy is bounded above by that of a normal vector with the same covariance matrix. So for all such vectors X and for $Y \sim N(0, \Sigma)$

$$h(X) \le h(Y).$$

Proposition 5.10. *The differential entropy of a random vector $Y \sim N(\mu, \Sigma)$, $Y \in \mathbb{R}^n$, is given by*

$$h(Y) = \frac{n}{2}\ln(2\pi e) + \frac{1}{2}\ln\det(e\Sigma).$$

Corollary 5.11. *For a normal random vector $Z \sim N(\mu, \sigma^2 I_n)$,*

$$h(z) = \frac{n}{2}\ln(2\pi e) + \frac{n}{2}\ln\det(e\sigma^2).$$

Proposition 5.12 (Binary KL Divergence is Upper Bounded by Square Distance)**.**

$$\frac{1}{1+e^{-x}}\ln\frac{1+e^{-y}}{1+e^{-x}} + \frac{1}{1+e^{x}}\ln\frac{1+e^{y}}{1+e^{x}} \leq \frac{(x-y)^2}{8}.$$

Remark. Note that this is the KL divergence between two binary random variables since we have

$$\frac{1}{1+e^{-x}} + \frac{1}{1+e^{x}} = \frac{e^x}{e^x+1} + \frac{1}{1+e^x}$$

and

$$\log\frac{1/1+e^{-x}}{1/1+e^{-y}} = \ln\frac{1+e^{-y}}{1+e^{-x}}.$$

Finally before reaching the main results, we state one more divergence inequality.

Definition 5.2 (Change of Measure)**.** Suppose that X is a random variables and F is some event. Let $f(x) = p(F|X = x)$. Suppose that Y is a random variable defined on the same space and taking on some subset of the range of X. Then

$$p(F|X \leftarrow Y)$$

denotes the conditional at the realisation of Y under the distribution conditioned on the X. In other words $p(F|X \leftarrow Y) = f(Y)$,

Proposition 5.13. *Suppose that $\theta : \Omega \to \Theta$ is a random variable with the properties previously specified and $\tilde{\theta} : \Omega \to \tilde{\Theta}$ is a random variable satisfying $Y_{t+1} \perp \tilde{\theta}|\theta, H_t$ for all $t \in \mathbb{Z}^+$ and $\tilde{\Theta} \subseteq \Theta$. Then*

$$\mathbb{E}\left[\mathrm{D_{KL}}\left(p(Y_{t+1}|\theta, H_t) || \, p(Y_{t+1}|\tilde{\theta}, H_t)\right)\right] \leq \mathbb{E}\left[\mathrm{D_{KL}}\left(p(Y_{t+1}|\theta, H_t) || \, p(Y_{t+1}|\theta \leftarrow \tilde{\theta}, H_t)\right)\right].$$

With these propositions in hand, we first upper bound the logistic regression rate distortion function.

Lemma 5.14. *Suppose that $((x_t, Y_{t+1}))_{t\in\mathbb{Z}}$ is a sequence of data generated by the logistic regression process, with the sample pairs i.i.d.. For all $d \in \mathbb{N}$ and for all $\varepsilon > 0$,*

$$H_{\varepsilon,T} \leq \frac{d}{2}\ln\left(1 + \frac{1}{8\varepsilon}\right).$$

Proof. We construct a random variable that is sufficiently close to θ to be contained in $\tilde{\Theta}_{\varepsilon,T}$, then show that the mutual information between it and θ is less than or equal to the given bound. Since $H_{\varepsilon,T}$ is the infimum over $\tilde{\Theta}_{\varepsilon,T}$, the result then follows.

Consider $\tilde{\theta} = \theta + Z$, where $Z \sim N(0, \frac{8\varepsilon I_d}{d})$. Notice that $\tilde{\theta}$ still satisfies all the properties detailed previously: it is independent of Y_{t+1} given H_t and θ and independent of X_t. Moreover $\tilde{\theta}$ lies in $\tilde{\Theta}_{\varepsilon,T}$:

$$
\begin{aligned}
\frac{I(H_T;\theta|\tilde{\theta})}{T} &\leq \mathbb{I}(Y_1;\theta|\tilde{\theta},X_0) && (a)\\
&= \mathbb{E}\left[\mathrm{D}_{\mathrm{KL}}\left(p(Y_1|\theta,X_0)||\ p(Y_1|\tilde{\theta},X_0)\right)\right] && (b)\\
&\leq \mathbb{E}\left[\mathrm{D}_{\mathrm{KL}}\left(p(Y_1|\theta,X_0)||\ p(Y_1|\theta\leftarrow\tilde{\theta},X_0)\right)\right] && (c)\\
&= \mathbb{E}\left[\frac{1}{1+e^{-\theta^TX_0}}\ln\frac{1+e^{-\tilde{\theta}^TX_0}}{1+e^{-\theta^TX_0}}+\frac{1}{1+e^{\theta^TX_0}}\ln\frac{1+e^{\tilde{\theta}^TX_0}}{1+e^{\theta^TX_0}}\right] && (d)\\
&\leq \frac{\mathbb{E}\left[(\theta^TX_0-\tilde{\theta}^TX_0)^2\right]}{8} && (e)\\
&= \frac{\mathbb{E}\left[||\tilde{\theta}-\theta||_2^2\right]}{8} && (f)\\
&= \varepsilon. && (g)
\end{aligned}
$$

Here (a) follows from Lemma 5.8, (b) from Lemma 2.6, (c) from Proposition 5.13, (d) by definition of change of measure, (e) from Proposition 5.12, (f) from the fact that $X_0 \sim N(0, I_d)$, meaning that we can write

$$
\begin{aligned}
\mathbb{E}\left[\left((\theta-\tilde{\theta})^TX_0\right)^2\right] &= \mathbb{E}\left[(\theta-\tilde{\theta})^TX_0X_0^T(\theta-\tilde{\theta})\right]\\
&= \mathbb{E}\left[\mathbb{E}\left[(\theta-\tilde{\theta})^TX_0X_0^T(\theta-\tilde{\theta})|\tilde{\theta},\theta\right]\right]\\
&= \mathbb{E}\left[(\theta-\tilde{\theta})^T(\theta-\tilde{\theta})\right],
\end{aligned}
$$

by the tower property of conditional expectation (Lemma A.8), and (g) by the construction of $\tilde{\theta}$.

Therefore $H_{\varepsilon,T} \leq \mathbb{I}(\theta, \tilde{\theta}$. We can calculate this by the design of $\tilde{\theta}$ and our bounds on differential entropy:

$$
\begin{aligned}
\mathbb{I}(\theta,\tilde{\theta}) &= h(\tilde{\theta}) - h(\tilde{\theta}|\theta)\\
&= h(\tilde{\theta}) - \frac{d}{2}\ln\left(2\pi e\frac{8\varepsilon}{d}\right) && (a)\\
&\leq \frac{d}{2}\ln\left(2\pi e\left(\frac{1}{d}+\frac{8\varepsilon}{d}\right)\right) - \frac{d}{2}\ln\left(2\pi e\frac{8\varepsilon}{d}\right) && (b)\\
&= \frac{d}{2}\ln\left(\frac{1}{8\varepsilon}+1\right),
\end{aligned}
$$

where (a) follows from Proposition 5.10 and (b) from Proposition 5.9 and Corollary 5.11, the definition of $\tilde{\theta}$ and the fact that variance of independent random variables is additive. □

Finally, we can use this to bound the optimal error.

Theorem 5.15. *For (X_t, Y_{t+1}) generated by the logistic regression process, for all $d \in \mathbb{Z}$*

and for all $T \in \mathbb{N}$,

$$\mathcal{L}_T \leq \frac{d}{2T}\left(1 + \ln(1 + \frac{T}{4d}\right).$$

Proof. We take the infimum over the rate-distortion bound:

$$\begin{aligned}
\mathcal{L}_T &\leq \inf_{\varepsilon \geq 0} \frac{H_{\varepsilon,T}(\theta)}{T} + \varepsilon && (a)\\
&\leq \inf_{\varepsilon \geq 0} \frac{d}{2T} \ln\left(\frac{1}{8\varepsilon} + 1\right) + \varepsilon && (b)\\
&\leq \inf_{\varepsilon \geq 0} \frac{d}{2T}\left(1 + \ln\left(\frac{1}{8\varepsilon} + 1\right)\right), && (c)
\end{aligned}$$

(a) following from Theorem 5.6, (b) from Lemma 5.14 and (c) from the choice of $\varepsilon = \frac{d}{2T}$.

Note that this final evaluation at $\varepsilon = \frac{d}{2T}$ is done to emphasise that loss decays in $\mathcal{O}(\frac{d}{T})$: it scales linearly with dimension and decays at a rate inversely proportional to the sample size. □

Corollary 5.16. *For any finite-dimensional logistic regression process, the underlying parameter* θ *and the corresponding conditional distribution* $p(Y_{t+1}|\theta, X_t)$ *can be learnt arbitrarily well.*

Chapter 6

Conclusion

In building up towards our theory of learning, we explored the nature of information and compression in great depth. We worked to stress the presence of Shannon entropy and mutual information as fundamental measures of complexity, especially through the Source Coding Theorems and the characterisation of the rate-distortion function. We showed the equivalence between compression and prediction, most clearly through the Kraft-McMillan Inequality. With the dual perspective this allowed us to take, we first showed that the Bayesian posterior is the unique optimal learner, then bounded the posterior by defining an appropriate rate-distortion function and viewing it as the optimal lossy compressor. This allowed us to derive an upper bound for the rate at which a logistic regression data-generating process could be optimally learnt. Through this we saw how the bounds allow us to determine how efficiently data from different sources can be learnt.

We did not explore here the exact bounds which have been derived ([10]) for more complex data-generating processes, such as neural networks. This would lead us into the domain where different theories' bounds diverge: other theoretical frameworks' results have proven to not be representative of empirical observations, as mentioned with agnostic PAC learnability, whereas the bounds of our information-based approach are still in agreement with observation. Exploring the bounds for both approaches would present an insight into why our focus on the complexity of the data-generating process is so valuable for understanding the true limits of learning.

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Appendix A

Measure-Theoretic Probability

We shall give a brief description of the construction of probability through measure theory. The aim of this is to provide a clearer understanding of some core results such as the weak law of large numbers and the tower property of conditional expectation. We aim to do so by the addition of a formal framework for probability. We include proofs only where relevant to understanding the nature of the objects defined, or to provide a refresher on measure-theoretic techniques. The discussion is adapted from [8].

A.1 Probability Spaces, Random Variables and Distributions

In probability theory, we look to model random events. We define events as subsets of a sample space Ω. Ω contains all possible outcomes of our random process. The particular collections of outcomes on which we focus is determined by the σ-algebra on Ω.

Definition A.1 (σ-Algebra)**.** A family $\mathcal{F}$ of subsets of Ω satisfying:

1. $A_i \in \mathcal{F}$ for all $i \in \mathbb{N} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$,
2. $A \in \mathcal{F} \implies A^c \in \mathcal{F}$,
3. $\Omega \in \mathcal{F}$.

defines a *σ-algebra* on Ω.

So each element $A \in \mathcal{F}$ denotes a specific event and $\mathcal{F}$ describes the set of events with which we are concerned as a whole.

By the given conditions, we have that $\emptyset \in \mathcal{F}$ as the complement of Ω and from this, we get that for events $A, B \in \mathcal{F}$, we can also discuss:

1. Finite unions: $A \cup B \in \mathcal{F}$ (take the union with $\emptyset$ for $i \geq 2$)
2. Finite intersections: $A \cap B \in \mathcal{F}$ ($A \cap B = (A^c \cup B^c)^c$).

To assign probabilities to random events, we define a measure on a set:

Definition A.2 (Measure)**.** A measure μ is a non-negative function mapping from a σ-algebra to $\mathbb{R}^+$,

$$\mu : \mathcal{F} \to [0, \infty)$$

satisfying:

1. σ-*additivity*: $\mu(\cup_i A_i) = \sum_i \mu(A_i)$ for pairwise disjoint sets (A_i).
2. $\mu(\emptyset) = 0$.

In general, we have the following:

Definition A.3 (Measurable and Measure Spaces)**.** A set equipped with a σ-algebra, $(\Omega, \mathcal{F})$ forms a *measurable* space. When a measure is defined on the space, the triple $(\Omega, \mathcal{F}, \mu)$ forms a *measure space.*

When dealing with random events, our restriction on μ is stronger:

Definition A.4 (Probability Measure)**.** A *probability measure* $P : \mathcal{F} \to [0, 1]$ defined on a measurable space is a measure satisfying the additional property $P(\Omega) = 1$.

This leads to the natural definition of a probability space:

Definition A.5 (Probability Space)**.** The triplet $(\Omega, \mathcal{F}, P)$ for P a probability measure defines a probability space.

Example A.1 (Measure Spaces)**.** Sample spaces can be any kind of space, however it is often a familiar one. On the real numbers, we often consider the probability triple $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$. Here $\mathcal{B}(\mathbb{R})$ refers to the Borel σ-algebra. One construction of this is as the smallest σ-algebra containing all open sets on the real line.

In particular, the Borel σ-algebra is the intersection of all possible σ-algebras containing the open intervals. This intersection is non-empty since the power set of $\mathbb{R}$ is a σ-algebra and contains all open sets.

The measure λ is the Lebesgue measure. For an interval $I = [a, b)$ or any such combination of open and closed ends, $\lambda(I) = b - a$.

The following features of the probability measure are useful to note, both to confirm that the constructions matches our intuition, and for use later on.

Lemma A.1 (Properties of the Probability Measure)**.** *For P a probability measure on some σ-algebra $\mathcal{F}$,*

1. *For $A, B \in \mathcal{F}$, $A \subseteq B \implies P(A) \leq P(B)$.*
2. *For a sequence $(A_n)_{n=1}^{\infty} \in \mathcal{F}$, such that*
$$A_1 \subset A_2 \subset \ldots \subset A_n \subset \ldots,$$
$P(\cup_{n=1}^{\infty} A_n) = \lim_{n\to\infty} P(A_n)$.
3. *For $(B_n)_{n=1}^{\infty} \in \mathcal{F}$ such that*
$$B_1 \supset \ldots \supset B_n \supset \ldots,$$
$P(\cap_{n=1}^{\infty} B_n) = \lim_{n\to\infty} P(B_n)$.

Proof. (1) follows by writing
$$P(B) = P(A) + P(B \setminus A)$$
and remembering that measures are nonnegative.

For (2), rewrite this union as

$$A_1 \cup (A_2 \setminus A_1) \cup \big(A_3 \setminus (A_1 \cup A_2)\big) \cup \dots$$

and denote this sequence $(C_n)_n$. These sets are now pairwise disjoint, and so by σ-additivity, we have

$$P(\cup A_n) = P(\cup C_n) = \lim_{n\to\infty} \sum_n P(C_n) = \lim_{n\to\infty} P(A_n).$$

(3) now follows since the complement of A_k has probability measure $1 - P(A_k)$ for all $k \in \mathbb{Z}$. Writing

$$P(\cap_{n=1}^{\infty} B_n) = 1 - P(\cup_{n=1}^{\infty} B_n^c) = 1 - \lim_{n\to\infty} P(B_n^c) = \lim_{n\to\infty} P(B_n),$$

we get the desired result.

Corollary A.2 (Union Bound)**.** *For $A_1, \dots, A_n \in \mathcal{F}$,*

$$P(\cup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} P(A_i)$$

Proof. As in the proof of property 2 above, we can write

$$\cup_{i=1}^{n} A_i = A_1 \cup (A_2 \setminus A_1) \cup \dots \cup (A_n \setminus (A_1 \cup \dots \cup A_{n-1})).$$

Then for each i, we have

$$A_i \setminus (A_1, \dots, A_{i-1}) \subseteq A_i.$$

Therefore applying property 1 and by σ-additivity, we get that

$$P(\cup_{i=1}^{n} A_i) = \sum_{i=1}^{n} P(A_i \setminus (A_1, \dots, A_{i-1}) \leq \sum_{i=1}^{n} P(A_i).$$

Probability spaces describe the behaviour of random variables. The definition of a random variable is initially surprising:

Definition A.6 (Measurable Function)**.** A function between σ-algebras

$$X : \mathcal{F} \to \mathcal{H}$$

is measurable if for all $H \in \mathcal{H},\ X^{-1}(H) \in \mathcal{F}$.

Definition A.7 (Random Variable)**.** A random variable is a measurable function X from a probability space $(\Omega, \mathcal{F}, P)$ to a measurable space $(H, \mathcal{H})$.

We can view a random variable as a function which maps each element from our abstract family of events $\mathcal{F}$, to a subset of the set of all values that X can take on, $\mathcal{H}$. Often H will be a space such as $\mathbb{R}$ or $\mathbb{R}^d$, but this need not be the case.

We are often interested in a particular random variable, and want to know the partition of events that the random variable variable creates. This can be described by the σ-algebra of the variable:

Definition A.8 (σ-Algebra Generated by a Random Variable)**.** For a random variable X mapping from a probability space $(\Omega, \mathcal{F}, P)$ to $(H, \mathcal{H})$, the σ-algebra generated by X is given by

$$\sigma(X) := \{X^{-1}(A) : A \in \mathcal{H}\}.$$

Note that $\sigma(X) \subseteq \mathcal{F}$ by (A.6).

Example A.2. Let Ω represent the space of all possible trajectories of lottery balls during a draw, and let H represent the space of all possible winning sequences. If the lottery drew 6 balls from the set $S = \{1, 2, 3, \ldots, 25\}$ with replacement, then H would be the set S^6. Let X be the random variable mapping a trajectory of the balls in 3D space to S^6.

A suitable σ-algebra $\mathcal{H}$ can be generated by X. All unions of possible winning sequences, all complements of such sequences and the event that any sequence is drawn would all be members of $\sigma(X)$, for example.

Moreover the preimage under X of the event $h \in \mathcal{H}$ that the winning numbers are either $(1, 2, 3, 5, 8, 13)$ or $(2, 3, 5, 7, 11, 13)$ would be an element of $\mathcal{F}$, the σ-algebra on Ω. The preimage would be the set of trajectories such that either of those pairs are chosen. As would the preimage of the event that the winning number is not $(1, 3, 6, 10, 15, 21)$, or the event that any element of S^6 wins. The preimage of this last example would contain every measurable set of $\mathcal{F}$, since the probability of the preimage must be 1.

We can also consider the combinations of events created by multiple random variables on the same space:

Definition A.9 (Product σ-Algebra)**.** Let X and Y be defined on $(\Omega, \mathcal{F}, P)$ and suppose that they generate σ-algebras $\mathcal{F}_1, \mathcal{F}_2$. Then $\mathcal{F}_1 \otimes \mathcal{F}_2$ denotes the smallest σ-algebra of subsets of Ω containing

$$\{A \times B : A \in \mathcal{F}_1, B \in \mathcal{F}_2\}.$$

This is known as the product σ-algebra and the space $(\Omega, \mathcal{F}_1 \otimes \mathcal{F}_2)$ is called the product probability space.

Definition A.10 (Probability Distribution)**.** For a random variable X, the distribution of X is a probability measure defined on H. It assigns to each Borel set a probability equal to that of the set's preimage under X:

$$Q_X(\mathcal{H}) := P(X^{-1}(\mathcal{H})) := P(X \in \mathcal{H}).$$

For a random variable taking values in $\mathbb{R}$, this becomes

$$Q_X(\mathcal{B}(\mathbb{R})) := P(X^{-1}(\mathcal{B}(\mathbb{R}))) = P(X \in \mathcal{B}(\mathbb{R})).$$

So when we write $P(X \geq 0)$ for some random variable X, we are more formally considering the probability measure on the preimage of $[0, \infty)$.

Example A.3 (Normal Distribution)**.** Consider a standard normally distributed random variable X defined on $(\Omega, \mathcal{F}, P)$ and taking values in $\mathbb{R}$. Then P satisfies that for all intervals $I = [a, b) \subseteq \mathbb{R}$,

$$P(X^{-1}(I)) = \int_a^b \frac{1}{\sqrt{2\pi}} e^{-x^2/2}\, dx.$$

A.2 Convergence, Expectation and Independence

We now define a collection of properties which characterise the behaviour of random variables. We begin with a definition of two notions of convergence in a general measure space and a discussion of their relationship in a probability space.

A.2.1 Convergence

Definition A.11. A sequence of functions f_n is said to converge *μ-almost everywhere* to an almost-everywhere finite measurable function f if there exists some set $\mathcal{A}$, $\mu(\mathcal{A}) = 0$, such that for all $x \in \mathcal{A}^c$,

$$\lim_{n\to\infty} f_n(x) = f(x).$$

We say that $f_n \to f$ μ-a.e.

Definition A.12. If μ is a probability measure, then we say that f_n *converges almost-surely* (*a.s.*) instead.

Definition A.13. A sequence of functions f_n is said to *converge in measure* μ if for all $\varepsilon > 0$, we have

$$\lim_{n\to\infty} \mu(|f_n - f| > \varepsilon) := \lim_{n\to\infty} \mu\big\{\omega \in \Omega : |f_n(\omega) - f(\omega)| > \varepsilon\big\} = 0.$$

Definition A.14. If μ is a probability measure, then we say that the random variables f_n *converge in probability* to f.

Since random variables are measurable functions, note that these notions of convergence apply for sequences of random variables, too.

Within a probability space, the notion of almost sure convergence is strictly stronger than that of convergence in probability:

Theorem A.3. *Consider a sequence of random variables $(X_n)_{n=1}^{\infty}$ defined on the probability space $(\Omega, \mathcal{F}, P)$. Suppose that (X_n) converges almost-surely to an almost-everywhere finite X. Then X_n converges to X in probability.*

Proof. We have for any $\varepsilon > 0$

$$\{\omega \in \Omega : |X_n(\omega) - X(\omega)| > \varepsilon\} \subseteq \cup_{k=n}^{\infty} \{\omega \in \Omega : |X_k(\omega) - X(\omega)| > \varepsilon\} := B_n.$$

Now (B_n) is a sequence of decreasing nested intervals. By A.1, we have

$$\lim_{n\to\infty} P(|X_n - X| > \varepsilon) \leq \lim_{n\to\infty} P(B_n) = P(\cap_{n=1}^{\infty} B_n) \leq P(X_n \not\to X) = 0,$$

since (X_n) converges almost-surely. (Note that the last inequality comes from those ω which do not converge pointwise to f but are ε-close eventually).

Remark. This theorem holds by the exact same proof for any sequence of functions converging almost-everywhere to an almost-everywhere finite function, provided that we are in a finite measure space: in other words, provided that $\mu(\Omega) < \infty$.

For spaces with infinite measure, the sequence (B_n) may be a sequence of sets with measure ∞. Part (3) of A.1 does not hold since we cannot write the intersection as the difference between the measure of the entire space and the measure of the union of the complements.

Example A.4 (Convergence in Probability does not Imply Convergence Almost Surely)**.** Consider the probability space given by $\Omega = [0,1]$, $\mathcal{F} = \mathcal{B}\big([0,1]\big)$ (the smallest σ-algebra containing all open sets in $[0,1]$), and $P = \lambda$, the Lebesgue measure.

We define the following sequence of random variables on it:

$$X_1(\omega) = \begin{cases} 1 & \omega \in [0,1/2] \\ 0 & \omega \in (1/2,1] \end{cases}, X_2(\omega) = \begin{cases} 1 & \omega \in (1/2,1] \\ 0 & \omega \in [0,1/2] \end{cases},$$
$$X_3(\omega) = \begin{cases} 1 & \omega \in [0,1/3] \\ 0 & \omega \in (1/3,1] \end{cases}, X_4(\omega) = \begin{cases} 1 & \omega \in (1/3,2/3) \\ 0 & \omega \in [0,1/3) \cup (2/3,1] \end{cases}, X_5(\omega) = \begin{cases} 1 & \omega \in (2/3,1] \\ 0 & \omega \in [0,2/3] \end{cases},$$

and so on. Now consider the limiting function $X(\omega) = 0$ uniformly on $[0,1]$. For all $\varepsilon > 0$, the sequence of measures $P(|X_n - X| > \varepsilon)$ converges to 0, so the sequence converges in probability to X. However for all $\omega \in \Omega$, there are infinitely many indices n such that $X_n(\omega) \neq 0$. Thus for no $\omega \in \Omega$ does $X_n(\omega) \to X(\omega)$. In particular, (X_n) does not converge almost-surely to X.

A.2.2 Expectation

We briefly outline the construction of the expectation of a real-valued random variable only. This follows a process commonly used in measure theory, where we define the desired property on simple objects first and then use this to recursively define it on more complex ones.

We begin by defining it for a simple random variable - one taking on finitely many values. We extend this to a non-negative random variable by taking a limit of the behaviour for simple non-negative random variables. Finally we extend to general random variables as the difference in expectation between $X^+ := \max(X,0)$ and $X^- := \min(0,-X)$, provided at least one is finite. Otherwise we say that the expectation does not exist. Note that the expectation of a random variable can be infinite and exist. This occurs when exactly one of X^+ or X^- is infinite.

A.2.3 Independence

We now expand back out to consider a general probability space $(\Omega, \mathcal{F}, P)$.

Definition A.15 (Independence of Events)**.** We define events A and B in $\mathcal{F}$ to be independent if
$$P(A \cap B) = P(A)P(B).$$

Corollary A.4. *The following are immediate consequences:*

1. *If $P(A) = 0$ or 1 then B is independent of A for all $B \in \mathcal{F}$.*

2. *if A and B are independent, then so are A^c and B*

Proof. First, if $P(A) = 0$ then so is $P(A \cap B)$ since $\{A \cap B\} \subseteq \{A\}$; if $P(A) = 1$ then $B \subseteq A$ so $\{A \cap B\} = \{B\}$ and $P(A \cap B) = P(B)$.

The second point follows by writing

$$\begin{aligned} P(A^c \cap B) &= P(B) - P(A \cap B) = P(B) - P(A)P(B) \\ &= (1 - P(A))P(B) = P(A^c)P(B). \end{aligned}$$

Suppose that we have different σ-algebras $\{\mathcal{F}_i\}$ defined on Ω. We define a notion of independence of σ-algebras. Equivalently, this is a notion of independence of sets of events: two σ-algebras are independent if the sets of events with which they are concerned do not influence each other. More precisely:

Definition A.16 (Independence of σ-Algebras)**.** For an at-most countable set of σ-algebras $\{\mathcal{F}_i\}$, we say that they are independent if the intersection of any permutation of events from a finite subset of the σ-algebras are independent: in other words, if

$$P(A_{i_1} \cap A_{i_2} \cap \ldots \cap A_{i_n}) = \Pi(a_{i_j}) \quad \text{for all } A_{i_j} \in F_{i_j} \subseteq \{\mathcal{F}_i\}, \text{ with } j \in \{1, \ldots, n\},\ n < \infty.$$

This in turn allows us to define the independence of random variables on a shared probability space.

Definition A.17 (Independence of Random Variables)**.** For a finite number of random variables, we say that they are independent if the σ-algebras generated by them are independent, where $\sigma(X_i) = \{X_i^{-1}(F) : F \in \mathbb{B}(\mathbb{R})\}$.

If the set of random variables is infinite, then they are independent if every finite subset of them is independent.

Remark. Note how we proceed here similarly to in our construction of the expectation of a random variable. We build up from events to σ-algebras to random variables.

Example A.5. Let us return to the lottery example described earlier. Suppose that we had three lottery balls for each number: one red, one blue, one green.

One set of events we might be interested in is the sequence of the numbers of the lottery balls as before. Another might be the sequence of the colours of the lottery balls. The random variables representing these events would both be defined on the same probability space: the space of all possible trajectories of the lottery balls. The σ-algebras generated by these random variables (as we detailed in the previous example in this setting) would be independent, since the numbers drawn are independent of the colours drawn. Note also that both of the σ-algebras generated by them would be subsets of the σ-algebra on Ω, $\mathcal{F}$.

A.3 Laws of Large Numbers

With the properties of convergence, expectation and independence now defined, we state the strong and weak laws of large numbers.

Theorem A.5 (Weak Law of Large Numbers)**.** *For a sequence (X_n) of independent and identically distributed real-valued random variables with finite mean,*

$$\frac{1}{n}\sum_{i=1}^{n} X_i \to E[X]$$

in probability.

Theorem A.6 (Strong Law of Large Numbers)**.** *For any sequence of i.i.d. real-valued random variables for which $E[X]$ exists,*

$$\frac{1}{n}\sum_{i=1}^{n} X_n \to E[X]$$

almost surely.

Proof. Omitted for both.

Let us translate the definitions of A.11 and A.13 to more intuitively explain the connection.

Our sample space Ω is now the space of n independent repetitions of the random event. Taking n to infinity, the weak law of large numbers states that for any fixed amount ε, the probability of the sample mean differing from the expected value of the random variable for any given sequence of repetitions goes to zero.

The strong law then states that the set of infinite sequences for which the average does not converge to the expected value of the random variable has probability measure zero.

By A.3 and A.4, we see that the strong law implies the weak law, but not vice versa.

A.4 Conditional Expectation and the Tower Property

Finally, we turn to the notion of conditional expectation, working up to the tower property of conditional expectation.

Let $(\Omega, \mathcal{F}, P)$ be a probability space and let $\mathcal{G} \subseteq \mathcal{F}$ also be a σ-algebra. (Note that $\mathcal{G} \subseteq \mathcal{F}$ means that all events in $\mathcal{G}$ are events in $\mathcal{F}$. Equivalently, $\mathcal{F}$ is a more fine-grained collection of relevant events than $\mathcal{G}$).

Definition A.18. A random variable $Y : \Omega \to \mathbb{R}$ on $(\Omega, \mathcal{F}, P)$ is $\mathcal{G}$-measurable if $Y^{-1}(F) \in \mathcal{G}$ for all $F \in \mathcal{B}(\mathbb{R})$.

Definition A.19. Suppose that a random variable X is $\mathcal{F}$-measurable and its expectation exists and is finite ($E|X| < \infty$). Suppose that there exists a random variable Y such that:

1. Y is $\mathcal{G}$-measurable.
2. $E[X\mathbb{1}_A] = E[Y\mathbb{1}_A]$ for all $A \in \mathcal{G}$.

Then Y is the conditional expectation of X with respect to $\mathcal{G}$, denoted $E[X|\mathcal{G}]$.

This definition immediately extends to conditioning on a random variable.

Definition A.20. Let X and Y be random variables defined on the same probability space $(\Omega, \mathcal{F}, P)$. Let $\mathcal{G} = \sigma(Y)$. By definition, Y is $\mathcal{G}$-measurable. Then

$$E[X|Y] := E[X|\sigma(Y)]$$

is the conditional expectation of X with respect to Y.

Example A.6. Suppose we are now observing a slightly different lottery. In this instance, there are only seven numbers, and the balls are ordered as follows: there are 7 balls with the number one, one for each colour of the rainbow; there are 6 with the number two, one for each colour of the rainbow except red; there are 5 with the number three, one for each colour except red and green, etc. 6 balls are drawn, with replacement.

As before, we could be interested in the two random variables mapping the trajectory of the balls to the sequence of lottery numbers and to the sequence of colours, respectively. In particular, we now focus on the conditional expectation for the sequence of numbers given the sequence of colours.

This function is a random variable depending on the colour-related event we specify: for example the expectation given all balls are red will be a sequences of ones. In that event that the sequence is all violet, then the expectation will be the average over all balls, since the number is uniformly distributed given that the colour is violet.

Let us now define the tower property of conditional expectation for first σ-algebras, before stating the consequent exact form for random variables.

One property of conditional expectation which we must note before proving the tower property is that it is unique. We omit the proof:

Lemma A.7. *If the conditional expectation exists, it is unique up to a set of measure zero.*

Theorem A.8 (Tower Property of Conditional Expectation)**.** *Suppose that X is a random variable on $(\Omega, \mathcal{F}, P)$ with finite expectation and let $\mathcal{G} \subseteq \mathcal{F}$ be a σ-algebra. Then the following properties hold:*

1. $E[E[X|\mathcal{G}]] = E[X]$.
2. *For* $\mathcal{G} \subseteq \mathcal{H} \subseteq \mathcal{F}$ *we have* $E[E[X|\mathcal{H}]|\mathcal{G}] = E[X|\mathcal{G}]$.

Proof. First, note that $\Omega \in \mathcal{G}$. Therefore we can write

$$E[X] = E[X\mathbb{1}_\Omega] = E\big[E[X|\mathcal{G}]\mathbb{1}_\Omega\big] = E\big[E[X|\mathcal{G}]\big].$$

We now use the uniqueness of conditional expectation to prove the second part.

First taking $E[X|\mathcal{H}]$ as our random variable, by the previous part, we have that for all $G \in \mathcal{G}$,

$$E\Bigg[E\big[E[X|\mathcal{H}]|\mathcal{G}\big]\mathbb{1}_G\Bigg] = E\Bigg[E[X|\mathcal{H}]\mathbb{1}_G\Bigg]$$

Note that since $\mathcal{G} \subseteq \mathcal{H}$,

$$G \in \mathcal{G} \implies G \in \mathcal{H}.$$

Therefore by definition, the conditional expectation of X given $\mathcal{H}$ satisfies

$$E\big[E[X|\mathcal{H}]\mathbb{1}_G\big] = E[X\mathbb{1}_G]$$

for all $G \in \mathcal{G}$.

By (A.7), $E\big[E[X|\mathcal{H}]|\mathcal{G}\big]$ is the conditional expectation of X given $\mathcal{G}$.

Corollary A.9 (Tower Property for Random Variables)**.** *Suppose as before that X is a random variable on $(\Omega, \mathcal{F}, P)$. Suppose that Y and Z are random variables defined on $(\Omega, \mathcal{F}, P)$, too. Then:*

1. $E\big[E[X|Y]\big] = E\big[E[X|\sigma(Y)]\big] = E[X]$.
2. $E\big[E[X|(Y,Z)]|Y\big] = E[X|Y]$.

Proof. The first part is an immediate application of (A.8). For the second part, the expectation given (Y, Z) is shorthand for conditioning on their product σ-algebra as in (A.9). The proof then follows exactly as above.